GREENAGRO BIOLOOP: AN AGRICULTURAL WASTE-TO-BIO-PRODUCT MANAGEMENT SYSTEM USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled "GreenAgro BioLoop: An Agricultural Waste-to-Bio-Product Management System using Zoho Creator" is the bonafide work of ASWINRAJA R 240701824, who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year [2026–2027], Semester [V], under my supervision.

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DECLARATION

I, Aswinraja R (240701824), hereby declare that the project report titled "GreenAgro BioLoop: An Agricultural Waste-to-Bio-Product Management System using Zoho Creator", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai

Date:

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ABSTRACT

Agricultural residues such as paddy straw, coconut husk and banana waste are often burnt or discarded, although they can be converted into compost, bio-fertiliser and biogas. Organisations that attempt this coordinate farmers, collection teams, processing centres, laboratories and customers through phone calls, paper registers and disconnected spreadsheets. The result is repeated data entry, no traceability from a product to its source waste, late updates of waste and stock status, and impact reports compiled by hand.

GreenAgro BioLoop is a Zoho Creator application that brings this chain into one system. Its data model has 18 forms connected by 21 lookup relationships, covering farmers, farms, waste, collection requests, schedules, collection, processing, quality tests, inventory, orders, distribution and environmental impact. It provides 33 reports (17 list, 15 kanban and one custom) and a dashboard of key indicators. Twelve Deluge scripts, run as eleven form workflows and one schedule, validate quantities, set defaults, mark waste as collected, release or exclude products after quality testing, compute order totals, reduce stock and send low-stock e-mail. Two Blueprints control the 14-stage processing batch and the 15-stage collection request. Seven permission profiles, including customer-portal profiles for farmers and buyers, control access.

The application was tested with 54 sample records through 22 test cases covering each functional requirement, and a review of the scripts and data produced a defect log with corrective actions. It is deployed [state the environments used and the production date]. Measured benefits: [state measured outcomes, for example the reduction in manual steps and report preparation time]. The project demonstrates how a low-code platform can deliver a traceable, role-based operations system for a circular-economy enterprise, and where careful testing is still required.

Keywords: *Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Blueprint; Role-Based Access Control; Agricultural Waste Management*

PROJECT AT A GLANCE

Project Title	GreenAgro BioLoop: An Agricultural Waste-to-Bio-Product Management System using Zoho Creator
Application Domain	Agriculture / Waste Management / Circular Economy
Student Name & Register No.	[] – [Register Number]
Zoho Creator Edition	[Education / Free / Standard / Professional]
Application Name (Link Name)	GreenAgro BioLoop ([greenagro-bioloop])
Published / Live URL	https://creatorapp.zoho.in/aswinraja.r.2024.cse_raja642/greenagro-bio-loop/#Page:DashBoard
Demo Credentials (per role)	[Admin: ... | Waste Collection Coordinator: ... | Farmer (portal): ...]
Number of Forms / Subforms	18 / 0
Number of Reports (by type)	List: 17 | Kanban: 15 | Calendar: 0 | Pivot/Chart: 0 | Others: 1 (custom report, All Farmers)
Number of Pages / Dashboards	1 (Dashboard: 4 KPI panels, 2 charts, status panel, embedded report)
Number of Workflows	Form: 11 | Schedule: 1 | Approval: 0 | Blueprint: 2 | Batch: 0
Deluge Functions / Custom Actions	12 scripts (DS-01 to DS-12); 0 custom functions
Integrations	Zoho Mail (sendmail); map link on address fields
Roles & Profiles Defined	Profiles: Administrator, Waste Collection Coordinator, Processing Center Manager, Environmental Compliance Officer, Inventory and Distribution Specialist, Farmer (portal), Bio-Product Buyer (portal), plus Read, Write, Developer, Customer. Role: CEO
Environments Used	[Development → Stage → Production]
Date of Production Deployment	

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DS	Deluge Script (script identifier)
ECO	Environmental Compliance Officer
ER	Entity–Relationship
FR	Functional Requirement
IDS	Inventory and Distribution Specialist
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PCM	Processing Center Manager
RBAC	Role-Based Access Control
REST	Representational State Transfer
UAT	User Acceptance Testing
UI	User Interface
WCC	Waste Collection Coordinator
Zia	Zoho’s AI assistant

CHAPTER 1 INTRODUCTION

1.1 Background

India generates several hundred million tonnes of crop residue every year. Paddy straw, coconut husk, banana stems and leaves, manure and similar by-products are often burnt in the open or left to rot, causing air pollution, loss of soil nutrients and a missed income opportunity [1]. The same residues can be converted into compost, bio-fertiliser, animal feed and biogas through composting, vermicomposting and anaerobic digestion.

GreenAgro BioLoop is modelled as a circular-economy enterprise that links farmers who generate agricultural waste with processing centres that convert it into saleable bio-products. The enterprise operates three processing centres (Chennai, Coimbatore and Trichy), a fleet of collection teams with vehicles, a catalogue of bio-products (organic compost, bio-fertiliser and biogas) and a customer base made up largely of the same farmers who supply the waste. A single unit of waste therefore passes through nine stages: generation, collection request, scheduling, collection, processing, quality testing, inventory, order and delivery, with an environmental-impact record kept for every processing batch.

Several departments take part in this flow: collection coordinators approve and schedule pickups, centre managers run processing batches, a compliance officer records carbon, water and land savings, and an inventory and distribution specialist manages stock and deliveries. Each of them needs to see the same, current data. This project builds that shared application on the Zoho Creator low-code platform.

1.2 Problem Statement

Collection, processing and sales of agricultural waste are coordinated today through phone calls, paper registers and disconnected spreadsheets. Because no single system links a waste record to its collection, processing batch, quality test, stock and customer order, the organisation cannot trace a product back to its source waste, cannot enforce business rules such as “do not collect more than what is available”, and cannot measure its environmental impact without a manual month-end exercise.

- **Data duplication and inconsistency:** the same farmer, farm and batch details are re-typed in several sheets.

- **No traceability:** a bag of compost cannot be traced to the farm, waste category, batch and test that produced it.
- **Late or missing status updates:** waste is still shown as available after it has been collected; stock is not reduced when an order is taken.
- **Quality gaps:** products can be sold without a recorded laboratory approval.
- **Manual reporting and no access control:** impact figures are compiled by hand, and every employee sees every sheet.

1.3 Objectives

- To design a normalised data model of 18 linked forms covering farmers, farms, waste, collection, processing, quality, inventory, orders, distribution and environmental impact.
- To automate status changes, validations, defaults and stock updates using 12 Deluge scripts attached to form and schedule workflows.
- To control the lifecycle of processing batches and collection requests using two Blueprints with 14 and 15 stages.
- To provide list and kanban reports for every form and a management dashboard of key indicators.
- To enforce role-based access through seven custom permission profiles, including customer-portal profiles for farmers and buyers.
- To deploy the application securely to a production environment with governance controls.

1.4 Scope of the Project

In scope: master data (farmers, farms, waste categories, processing centres, processing methods, collection teams, bio-products); transactional processes (waste records, collection requests, schedules, collection records, processing batches and records, quality tests, inventory, orders, distribution, environmental impact); 33 reports; one dashboard page; form and schedule workflows; two Blueprints; role-based access for internal users and customer-portal users; access through the web and the Zoho Creator mobile app.

Out of scope: online payment collection, GPS tracking of vehicles in real time, integration with external weather or mapping services, accounting and invoicing, and

any hardware integration such as sensors on digesters. These are listed as future enhancements in Section 9.2.

1.5 Justification for a Low-Code Approach

The business problem is data-centric and workflow-centric: many related forms, status-driven processes, validations, role-based views and simple dashboards. These are exactly the capabilities that a low-code application platform provides out of the box, so that development effort goes into modelling the domain rather than into infrastructure [2], [3]. Low-code platforms are also recognised as an enabler for small and medium enterprises that lack large software teams [4], and as a practical meeting point between model-driven engineering and day-to-day business development [5].

For GreenAgro BioLoop specifically, Zoho Creator offers: (i) built-in hosting, database, authentication and mobile apps, so no servers are managed; (ii) drag-and-drop form and report builders, which allowed 18 forms and 33 reports to be produced quickly; (iii) Blueprint for stage-wise process control; (iv) Deluge scripting for the business rules that configuration alone cannot express; and (v) customer-portal licensing for farmers and buyers. The result can be maintained by a business analyst or "citizen developer" after hand-over.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5 details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

In the existing (as-is) arrangement, a farmer who has crop residue to dispose of telephones or messages a collector. The collector notes the request in a paper register and verbally assigns a vehicle team. The team's route and fuel are planned by hand. After pickup, the weight and a quality remark are written on a slip and later copied into a spreadsheet. At the processing centre, each batch is tracked in a separate sheet; laboratory results arrive on paper or by e-mail; stock is counted manually; product orders are taken over the phone; and the environmental benefit (CO_2, water and land saved) is compiled at the end of the month. Figure 2.1 shows this flow.

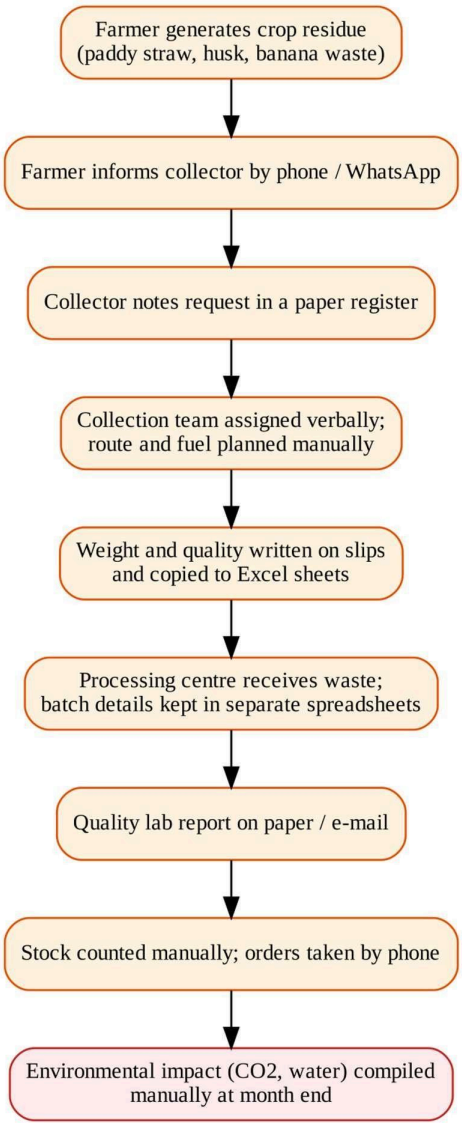


Figure 2.1: As-Is Process Flow of the Existing System

2.2 Limitations of the Existing System

- Information is re-entered at every hand-off, so errors and mismatches creep in between the register, the spreadsheets and the lab reports.
- Waste status, stock levels and order status are updated by memory, not by rules; a rejected product can still be sold.
- No traceability from a finished product back to the farm, waste category, collection and batch that produced it.
- Management cannot see live figures such as total orders, waste received and reused, or average processing efficiency.

- There is no role-based access; farmers and buyers cannot view their own orders or requests online.

2.3 Low-Code versus Traditional Development

Table 2.1 compares the two approaches for this particular application. The effort figures are planning estimates for a project of this size (18 forms, 33 reports, 12 scripts, 2 Blueprints, 1 dashboard).

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	Estimated 8–10 weeks: schema design, REST back-end, web UI, role management, testing	Estimated 2–3 weeks: forms, reports, workflows and Deluge configured on the platform
Skills required	Back-end language, database, front-end framework, DevOps	Data modelling, Deluge scripting, platform configuration
Hosting & infrastructure	Server, database and SSL certificate to be provisioned and maintained	Hosted by Zoho; no servers to manage
Mobile application	Separate Android/iOS app or responsive redesign	Zoho Creator mobile app available automatically
Security & access control	Authentication and role logic coded by hand	Profiles, roles, portal users and permissions configured
Maintenance & change	Code change, build and redeploy for each new field or report	Add a field or report in the builder and publish
Customisation limits	Unlimited, subject to effort	Bound by platform features, Deluge limits and UI components
Cost	Developer salaries plus hosting	Per-user subscription (Zoho Creator edition in use: [Education / Free / Standard / Professional])

2.4 Literature Survey

Seven sources were reviewed: the prescribed course text, vendor documentation, two peer-reviewed studies on low-code, an analyst report on the platform market, and a national policy document for the agricultural-waste domain. They are summarised in Table 2.2 and listed in the References.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	W. Rosam and O. Kalnichevski [2]	Low-Code Development: A Practical Guide	Explains low-code principles, platform architecture and the move from citizen developer to professional developer	Course text; basis for the justification in Section 1.5 and the spectrum analysis in Section 8.5
2	Zoho Corporation [3], [8]	Zoho Creator Help; Deluge Script Reference	Describes forms, reports, pages, workflows, Blueprint, profiles and the Deluge language	Primary reference for every feature used in Chapter 5
3	R. Sanchis et al. [4], 2020	Low-Code as Enabler of Digital Transformation in Manufacturing Industry	Low-code lets organisations with limited IT staff build operational applications quickly	Supports the choice of low-code for a small enterprise with several departments
4	D. Di Ruscio et al. [5], 2022	Low-code development and model-driven engineering: two sides of the same coin?	Relates low-code platforms to model-driven engineering and notes risks such as vendor lock-in	Informs the data-model-first approach (Section 4.2) and the limitations in Section 8.4
5	A. Bucaioni et al. [6], 2022	Modelling in low-code development: a multi-vocal systematic review	Surveys how low-code platforms support visual modelling and the open challenges	Context for the form, ER and Blueprint models in Chapter 4
6	Gartner, Inc. [7]	Magic Quadrant for Enterprise Low-Code Application Platforms	Defines evaluation criteria for low-code platforms: app building, governance, AI and integration	Criteria reused in the analysis of Section 8.4
7	Ministry of Agriculture & Farmers Welfare, Govt. of India [1], 2014	National Policy for Management of Crop Residues	Promotes in-situ and ex-situ use of crop residue instead of burning	Domain justification for collecting and processing residues into bio-products

2.5 Emerging Trends in Low-Code Development

Three trends shaped the design. First, **citizen development**: business users increasingly build and maintain their own applications, so the data model and process rules were kept simple and named in plain business language. Second, **AI-assisted building**: platforms now offer AI assistants for generating forms and Deluge code; where such help is used it must be reviewed and tested, a point discussed in Sections 5.9 and 6.4. Third, **hyper-automation and composable applications**: processes that were manual are connected by rules, and small applications are linked with other services through APIs. GreenAgro BioLoop applies the first two directly and prepares for the third by keeping each process in its own form and workflow.

2.6 Proposed System

The proposed system is a single Zoho Creator application, GreenAgro BioLoop, in which every stage of the waste-to-product chain is a form linked to its predecessor by lookup fields (Section 4.2). It overcomes the limitations of Section 2.2 as follows: data is entered once and reused through lookups; Deluge workflows change statuses, default values and stock levels automatically; the lifecycle of batches and requests is governed by Blueprints; the dashboard shows live totals; and seven permission profiles, including customer-portal profiles, control who may see and change what.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Administrator	Owner of the application	Manage users, profiles and settings; view and edit all data; receive low-stock e-mails
Waste Collection Coordinator	Operations staff	Review and approve collection requests by priority; schedule teams; record collections
Processing Center Manager	Plant manager	Create and run processing batches; maintain methods and centres; record processing results; log quality tests
Environmental Compliance Officer	Sustainability officer	Record CO_2, water and land savings; review quality tests
Inventory and Distribution Specialist	Stores and logistics	Maintain bio-product catalogue and stock; process orders; manage distribution
Farmer (Customer Portal)	Waste supplier and customer	Maintain own farm details; raise collection requests; place and track orders
Bio-Product Buyer (Customer Portal)	External purchaser	View product specifications; place and track orders

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall maintain master data for farmers and their farms (area, crop, soil, irrigation, land-deed number).	Admin, Farmer	High
FR-02	The system shall record agricultural waste by farmer, farm, category, generated and available quantity, date, location and status.	Admin, Farmer	High
FR-03	The system shall allow a collection request to be raised against a waste record and shall reject a request whose quantity exceeds the available quantity; new requests shall default to Pending.	WCC, Farmer	High
FR-04	The system shall move collection requests and processing batches through defined stages using Blueprint.	WCC, PCM	High
FR-05	The system shall schedule a collection (team, date, time, route) and record the collection; the linked waste record shall be marked Collected automatically.	WCC	High
FR-06	The system shall create a processing batch linked to a collection record, a processing centre and a processing method, and record its outcome (output, residue, efficiency, grade,	PCM	High

	energy).		

FR ID	Requirement	Role(s)	Priority
FR-07	The system shall record product quality tests, default a new test to Pending, make the product available on approval and remove its stock on rejection.	PCM, ECO	High
FR-08	The system shall maintain the bio-product catalogue and inventory by centre, and warn when stock falls below the reorder level (on entry and by scheduled e-mail).	IDS, Admin	Medium
FR-09	The system shall accept product orders only for approved products with sufficient stock, compute the order total, and reduce stock after the order is saved.	IDS, Farmer, Buyer	High
FR-10	The system shall record dispatch and delivery of each order (dates, quantity, status, driver, vehicle, proof).	IDS	Medium
FR-11	The system shall record environmental impact per batch: waste received and reused, reuse percentage, CO_2, water and land saved, and verification status.	ECO	Medium
FR-12	The system shall present a dashboard of total order amount, average processing efficiency, total waste received and reused, waste-status counts and stock and request charts.	Admin	Medium
FR-13	The system shall provide list and kanban reports for each form.	All	Medium
FR-14	The system shall restrict each user to the forms and reports permitted by the assigned permission profile.	All	High

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	List and kanban reports with up to [N] records shall load within [N] seconds on a broadband connection.
NFR-02	Security	Users shall access only the forms and reports permitted by their permission profile; portal users shall be separated from internal users.
NFR-03	Usability	The application shall be usable in a web browser and in the Zoho Creator mobile app; field labels shall use business terms.
NFR-04	Availability	The application shall be hosted on Zoho's cloud infrastructure, so no application server is maintained by the organisation.
NFR-05	Maintainability	Forms, reports and scripts shall follow a consistent naming convention so that a non-programmer can modify them.
NFR-06	Auditability	Changes to records shall be traceable through the Zoho Creator audit trail.
NFR-07	Data integrity	Relationships between forms shall be enforced through lookup fields; business rules shall be enforced by validation scripts.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator [Education / Professional] Edition

Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox (latest version)

Category	Specification
API testing tool	Postman (not required by the current scope; no external API calls are used)
Other Zoho services	Zoho Mail (e-mail alert through the Deluge sendmail task)
Mobile	Zoho Creator mobile app (Android / iOS)
Application settings	Date format dd-MMM-yyyy; time zone Asia/Kolkata; time format 24-hour
Client hardware	Intel Core i3/i5 or above, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the actors and the principal use cases. Three of them are described in detail in Tables 3.5 to 3.7.

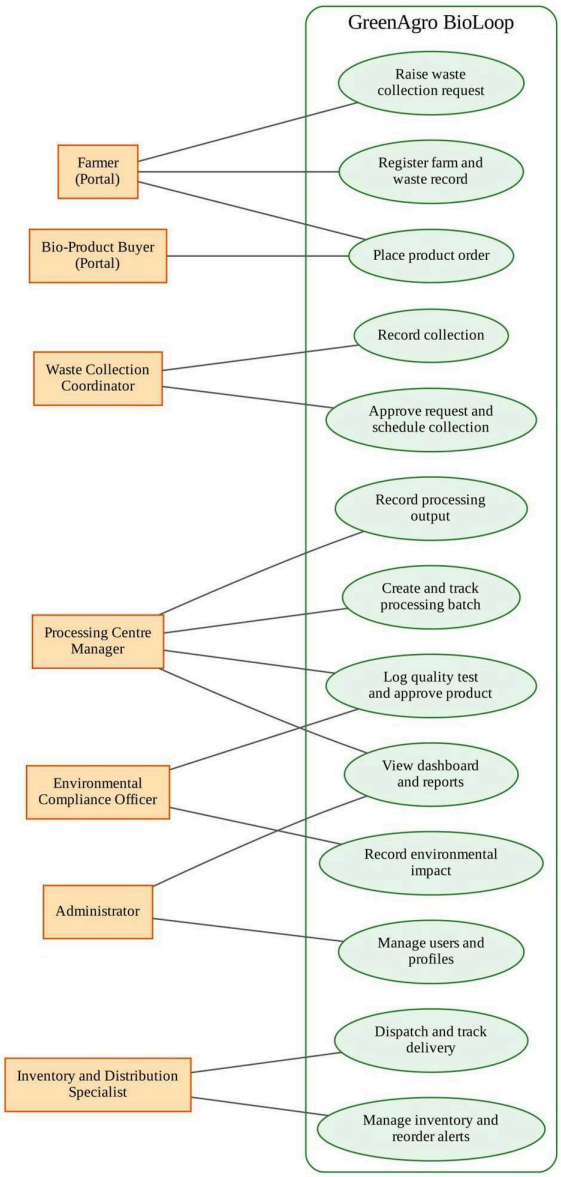


Figure 3.1: Use Case Diagram of the Application

Table 3.5: Use Case Description – Raise a Waste Collection Request

Use Case ID	UC-01
Use Case Name	Raise a Waste Collection Request
Actor(s)	Farmer (portal) or Waste Collection Coordinator
Pre-condition	A waste record exists with available quantity greater than zero.
Main Flow	1. Open the Waste Collection Requests form. 2. Select the waste record; the status defaults to Pending (DS-07). 3. Enter requested quantity, collection address, date, priority and instructions. 4. Submit; DS-02 compares the requested quantity with the available quantity. 5. The request enters the Blueprint at “Request Created”.

Alternate Flow	If requested quantity exceeds available quantity, an error message is shown and the record is not saved. The user corrects the quantity and resubmits.
Post-condition	A request with status Pending is stored and visible in the request reports.
Related FR	FR-03, FR-04

Table 3.6: Use Case Description – Approve a Quality Test and Release the Product

Use Case ID	UC-02
Use Case Name	Approve a Quality Test and Release the Product
Actor(s)	Processing Center Manager, Environmental Compliance Officer
Pre-condition	A processing batch has been completed and a quality test record exists for the batch and product.
Main Flow	1. Open the Product Quality Tests record (result defaults to Pending, DS-04). 2. Enter test parameters, lab and technician; set the test result. 3. Change Approval Status to Approved. 4. DS-05 checks the inventory for the product and centre and creates a row if none exists. 5. The product becomes eligible for ordering (DS-08).
Alternate Flow	If Approval Status is set to Rejected, DS-03 sets the product's available stock to zero, appends a rejection note and marks a Started batch as Failed.
Post-condition	Approved product can be ordered; rejected product cannot be sold from stock.
Related FR	FR-07

Table 3.7: Use Case Description – Place a Product Order

Use Case ID	UC-03
Use Case Name	Place a Product Order
Actor(s)	Farmer or Bio-Product Buyer (portal), Inventory and Distribution Specialist
Pre-condition	The product has an approved quality test and an inventory row with stock.
Main Flow	1. Open the Product Orders form and select farmer and product. 2. Enter the order quantity. 3. On submit, DS-08 checks approval and stock and calculates total amount = quantity × selling price. 4. After saving, DS-09 reduces the available stock. 5. Dispatch is recorded later in Product Distribution.
Alternate Flow	If stock is insufficient or the product is not approved, a message is displayed (see defect BUG-02 in Table 6.2 for the current behaviour).
Post-condition	Order saved with total amount; stock reduced.
Related FR	FR-09, FR-10

CHAPTER 4 SYSTEM DESIGN

4.1 System Architecture

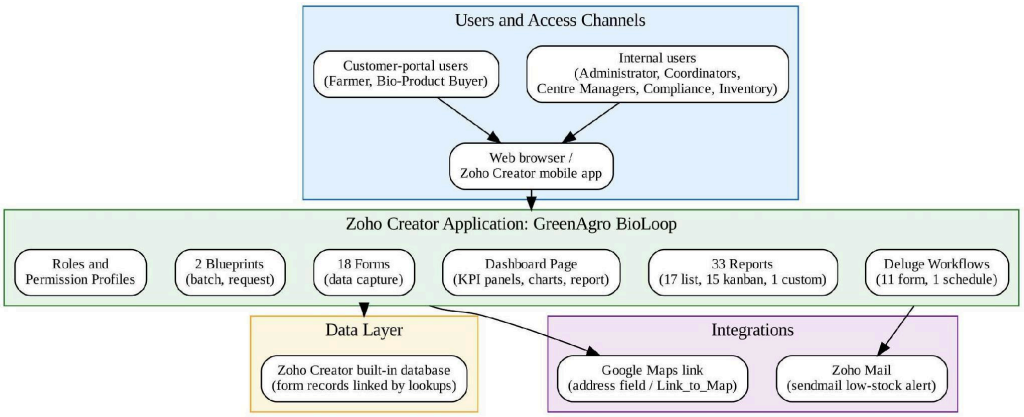


Figure 4.1: System Architecture of the Application

The solution has four layers. The **access layer** consists of internal users and customer-portal users who reach the application through a web browser or the Zoho Creator mobile app. The **application layer** is the GreenAgro BioLoop application itself, made of 18 forms, 33 reports, one dashboard page, 12 Deluge-based workflows (11 form workflows and one schedule) and two Blueprints; access to each component is governed by permission profiles and the CEO role. The **data layer** is the built-in Zoho Creator database, where every form is a table and lookup fields hold the relationships. The **integration layer** is deliberately thin in this version: the Deluge sendmail task uses Zoho Mail for low-stock alerts, and address fields offer a map link.

4.2 Data Model

The data model follows the physical flow of material. Master data (farmers, farms, waste categories, processing centres, processing methods, collection teams and bio-products) is separated from transactional data so that it is entered once and reused. Each transactional form refers to its predecessor through a lookup field: a waste record refers to a farmer, farm and category; a collection request refers to a waste record; a schedule refers to a request and a team; a collection record refers to a schedule and a waste record; a processing batch refers to a collection record, a centre and a method; and so on to orders and distribution. There are 21 lookup relationships in total (Figure 4.2, Table 4.1). Subforms were not required because no form owns a repeating group of

child rows; each child is a first-class record with its own lifecycle.

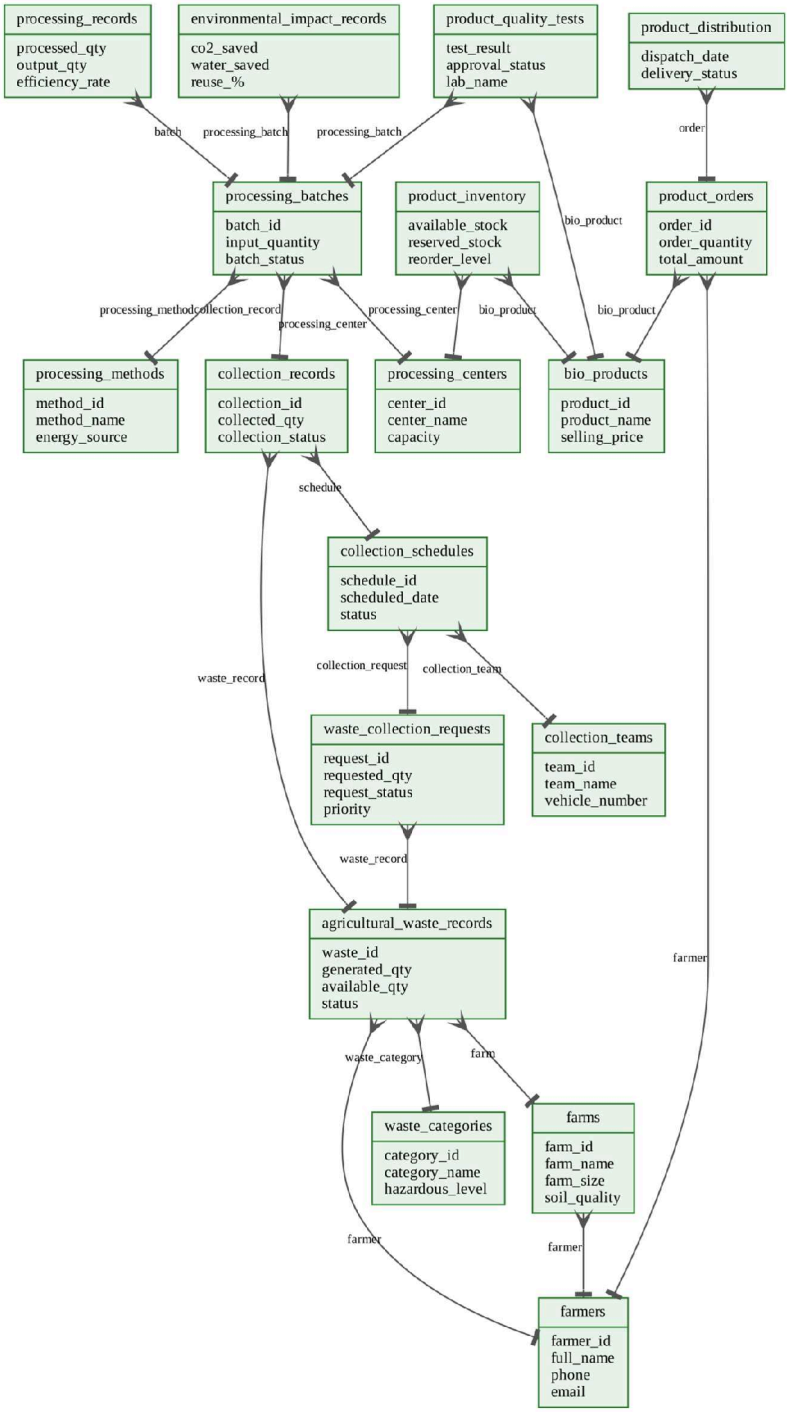


Figure 4.2: Entity–Relationship Diagram

Table 4.1: Entities and Relationships

Entity (Form)	Related To	Relationship Type	Cardinality	Implemented Using
Farms	Farmers	Many-to-One	M : 1	Lookup field “farmer” in Farms

Entity (Form)	Related To	Relationship Type	Cardinality	Implemented Using
Agricultural Waste Records	Farmers	Many-to-One	M : 1	Lookup field “farmer” in Agricultural Waste Records
Agricultural Waste Records	Farms	Many-to-One	M : 1	Lookup field “farm” in Agricultural Waste Records
Agricultural Waste Records	Waste Categories	Many-to-One	M : 1	Lookup field “waste_category” in Agricultural Waste Records
Waste Collection Requests	Agricultural Waste Records	Many-to-One	M : 1	Lookup field “waste_record” in Waste Collection Requests
Collection Schedules	Waste Collection Requests	Many-to-One	M : 1	Lookup field “collection_request” in Collection Schedules
Collection Schedules	Collection Teams	Many-to-One	M : 1	Lookup field “collection_team” in Collection Schedules
Collection Records	Collection Schedules	Many-to-One	M : 1	Lookup field “schedule” in Collection Records
Collection Records	Agricultural Waste Records	Many-to-One	M : 1	Lookup field “waste_record” in Collection Records
Processing Batches	Collection Records	Many-to-One	M : 1	Lookup field “collection_record” in Processing Batches
Processing Batches	Processing Centers	Many-to-One	M : 1	Lookup field “processing_center” in Processing Batches
Processing Batches	Processing Methods	Many-to-One	M : 1	Lookup field “processing_method ” in Processing Batches
Processing Records	Processing Batches	Many-to-One	M : 1	Lookup field “batch” in Processing Records
Environmental Impact Records	Processing Batches	Many-to-One	M : 1	Lookup field “processing_batch” in Environmental Impact Records
Product Quality Tests	Processing Batches	Many-to-One	M : 1	Lookup field “processing_batch” in Product Quality Tests

Product Quality Tests	Bio Products	Many-to-One	M : 1	Lookup field “bio_product” in Product Quality Tests

Entity (Form)	Related To	Relationship Type	Cardinality	Implemented Using
Product Inventory	Bio Products	Many-to-One	M : 1	Lookup field “bio_product” in Product Inventory
Product Inventory	Processing Centers	Many-to-One	M : 1	Lookup field “processing_center” in Product Inventory
Product Orders	Farmers	Many-to-One	M : 1	Lookup field “farmer” in Product Orders
Product Orders	Bio Products	Many-to-One	M : 1	Lookup field “bio_product” in Product Orders
Product Distribution	Product Orders	Many-to-One	M : 1	Lookup field “order” in Product Distribution

4.3 Form Design

One field-specification table is given for each of the 18 forms (Tables 4.2 onwards). Field types are named as on the Zoho Creator platform: a “Drop Down” holds a fixed list of choices, whereas a “Lookup” takes its values from another form. No field is marked mandatory in the current build (see Section 8.4); validation is performed by Deluge scripts at submission. Every form uses left-aligned labels, a success message of the form “<Form> Added Successfully”, Submit and Reset buttons on add, and Update and Cancel buttons on edit.

Table 4.2: Field Specification – Agricultural Waste Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Waste Id	waste_id	Auto Number	—	System generated, starts at 1	
Farmer	farmer	Lookup	No	Values from Farmers; sorted ascending; “Add New” enabled	
Farm	farm	Lookup	No	Values from Farms; sorted ascending; “Add New” enabled	
Waste Category	waste_category	Lookup	No	Values from Waste Categories; sorted ascending; “Add New” enabled	
Generated Quantity	generated_quantity	Number	No		
Available Quantity	available_quantity	Number	No		Compared by DS-02

Generation Date	generation_date	Date	No		
Location	location	Address	No		Address with map link and coordinates

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Status	status	Drop Down	No	Available / Collected / Reserved / Processed (Others allowed)	Set to Collected by DS-01; kanban column
Weather Condition	weather_condition	Single Line	No		
Remarks	remarks	Multi Line	No		

Table 4.3: Field Specification – Bio Products Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Product Id	product_id	Auto Number	—	System generated, starts at 1	
Product Name	product_name	Single Line	No		
Product Type	product_type	Drop Down	No	Compost / Fuel / Fertilizer / Feed (Others allowed)	
Description	description	Multi Line	No		
Unit	unit	Drop Down	No	Kilogram / Tonne / Bag / Box (Others allowed)	
Selling Price	selling_price	Currency (INR)	No		
Shelf Life	shelf_life	Number	No		
Storage Condition	storage_condition	Single Line	No		
Manufacturer	manufacturer	Single Line	No		
Certification	certification	Single Line	No		

Table 4.4: Field Specification – Collection Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Collection Id	collection_id	Auto Number	—	System generated, starts at 1	
Schedule	schedule	Lookup	No	Values from Collection Schedules; sorted ascending; “Add New” enabled	
Waste Record	waste_record	Lookup	No	Values from Agricultural Waste Records; sorted ascending; “Add New” enabled	Status set by DS-01
Collected Quantity	collected_quantity	Number	No		
Collection Date	collection_date	Date	No		

Collection Status	collection_status	Drop Down	No	Partial / Failed / Collected (Others allowed)	
Actual Weight	actual_weight	Number	No		
Quality Check	quality_check	Drop Down	No	Pass / Review / Fail (Others allowed)	
Collector Notes	collector_notes	Multi Line	No		
Gps Coordinates	gps_coordinates	Single Line	No		

Table 4.5: Field Specification – Collection Schedules Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Schedule Id	schedule_id	Auto Number	—	System generated, starts at 1	
Collection Request	collection_request	Lookup	No	Values from Waste Collection Requests; sorted ascending; “Add New” enabled	
Collection Team	collection_team	Lookup	No	Values from Collection Teams; sorted ascending; “Add New” enabled	
Scheduled Date	scheduled_date	Date	No		
Scheduled Time	scheduled_time	Time	No		
Status	status	Drop Down	No	Scheduled / Completed / In Progress / Cancelled (Others allowed)	
Route Plan	route_plan	Multi Line	No		
Driver Name	driver_name	Name	No		
Fuel Level	fuel_level	Number	No		
Weather Alert	weather_alert	Single Line	No		

Table 4.6: Field Specification – Collection Teams Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Team Id	team_id	Auto Number	—	System generated, starts at 1	
Team Name	team_name	Single Line	No		
Team Leader	team_leader	Name	No		
Phone Number	phone_number	Phone	No		
Vehicle Details	vehicle_details	Single Line	No		
Vehicle Number	vehicle_number	Single Line	No		
Team Availability	team_availability	Drop Down	No	On Duty / Available / Off Duty (Others allowed)	
Max Capacity	max_capacity	Number	No		
Assigned District	assigned_district	Single Line	No		
License Expiry	license_expiry	Date	No		

Table 4.7: Field Specification – Environmental Impact Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks

Processing Batch	processing_batch	Lookup	No	Values from Processing Batches; sorted ascending; “Add New” enabled	

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Total Waste Received	total_waste_received	Number	No		
Total Waste Reused	total_waste_reused	Number	No		
Waste Reuse Percentage	waste_reuse_percentage	Percent	No		
Reporting Date	reporting_date	Date	No		
Co2 Saved	co2_saved	Number	No		
Water Saved	water_saved	Number	No		
Land Saved	land_saved	Number	No		
Reporter Name	reporter_name	Name	No		
Verification Status	verification_status	Radio	No	Verified / Unverified (Others allowed)	

Table 4.8: Field Specification – Farmers Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Farmer Id	farmer_id	Auto Number	—	System generated, starts at 1	
Full Name	full_name	Name	No		
Phone Number	phone_number	Phone	No		
Email	email	Email	No		
Address	address	Address	No		
Registration Date	registration_date	Date	No		
Is Active	is_active	Radio	No	Active / Inactive (Others allowed)	
Primary Crop	primary_crop	Single Line	No		
Emergency Contact	emergency_contact	Phone	No		

Table 4.9: Field Specification – Farms Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Farm Id	farm_id	Auto Number	—	System generated, starts at 1	
Farm Name	farm_name	Single Line	No		
Farmer	farmer	Lookup	No	Values from Farmers; sorted ascending; “Add New” enabled	

Location	location	Address	No		
Farm Size	farm_size	Number	No		
Crop Type	crop_type	Single Line	No		
Soil Quality	soil_quality	Drop Down	No	Clay / Loamy / Silt / Sandy (Others allowed)	

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Irrigation Source	irrigation_source	Drop Down	No	Rain / Canal / Well / River (Others allowed)	
Land Deed Number	land_deed_number	Single Line	No		
Last Survey Date	last_survey_date	Date	No		

Table 4.10: Field Specification – Processing Batches Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Batch Id	batch_id	Auto Number	—	System generated, starts at 1	
Collection Record	collection_record	Lookup	No	Values from Collection Records; sorted ascending; “Add New” enabled	
Processing Center	processing_center	Lookup	No	Values from Processing Centers; sorted ascending; “Add New” enabled	
Processing Method	processing_method	Lookup	No	Values from Processing Methods; sorted ascending; “Add New” enabled	
Input Quantity	input_quantity	Number	No		
Start Date	start_date	Date	No		
Expected Completion Date	expected_completion_date	Date	No		
Batch Status	batch_status	Drop Down	No	Started / Failed / Completed / In Progress (Others allowed)	Set to Failed by DS-03; kanban column
Operator Name	operator_name	Name	No		
Batch Notes	batch_notes	Multi Line	No		

Table 4.11: Field Specification – Processing Centers Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Center Id	center_id	Auto Number	—	System generated, starts at 1	
Center Name	center_name	Single Line	No		
Address	address	Address	No		
Manager Name	manager_name	Name	No		

Phone Number	phone_number	Phone	No		
Processing Capacity	processing_capacity	Number	No		
Operating Hours	operating_hours	Single Line	No		
Certification Level	certification_level	Drop Down	No	ISO / National / Local (Others allowed)	
Last Inspection	last_inspection	Date	No		

Table 4.12: Field Specification – Processing Methods Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Method Id	method_id	Auto Numbe r	—	System generated, starts at 1	
Method Name	method_name	Single Line	No		
Description	description	Multi Line	No		
Suitable Waste Types	suitable_waste_types	Multi Select	No	Leaves / Manure / Husks / Stalks / Straw	
Expected Output Type	expected_output_type	Single Line	No		
Temperature Range	temperature_range	Single Line	No		
Duration Hours	duration_hours	Number	No		
Energy Source	energy_source	Drop Down	No	Solar / Electric / Gas / Biomass (Others allowed)	

Table 4.13: Field Specification – Processing Records Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Batch	batch	Lookup	No	Values from Processing Batches; sorted ascending; “Add New” enabled	
Processed Quantity	processed_quantity	Number	No		
Output Quantity	output_quantity	Number	No		
Waste Residue Quantity	waste_residue_quantity	Number	No		
Completion Date	completion_date	Date	No		
Processing Status	processing_status	Drop Down	No	Partial / Failed / Completed (Others allowed)	
Efficiency Rate	efficiency_rate	Percent	No		
Quality Grade	quality_grade	Drop Down	No	Grade A / Grade B / Grade C (Others allowed)	
Residue Disposal	residue_disposal	Single Line	No		
Energy Consumed	energy_consumed	Number	No		

Table 4.14: Field Specification – Product Distribution Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Order	order	Lookup	No	Values from Product Orders; sorted ascending; “Add New” enabled	
Delivery Address	delivery_address	Address	No		
Dispatch Date	dispatch_date	Date	No		
Expected Delivery Date	expected_delivery_date	Date	No		

Delivered Quantity	delivered_quantity	Number	No		

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Delivery Status	delivery_status	Drop Down	No	Returned / Dispatched / Delivered / In Transit (Others allowed)	
Driver Name	driver_name	Name	No		
Vehicle Number	vehicle_number	Single Line	No		
Proof Of Delivery	proof_of_delivery	URL	No		
Customer Signature	customer_signature	Single Line	No		

Table 4.15: Field Specification – Product Inventory Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Bio Product	bio_product	Lookup	No	Values from Bio Products; sorted ascending; “Add New” enabled	
Processing Center	processing_center	Lookup	No	Values from Processing Centers; sorted ascending; “Add New” enabled	
Available Stock	available_stock	Number	No		Alert by DS-11 and DS-12; reduced by DS-09
Reserved Stock	reserved_stock	Number	No		
Last Updated Date	last_updated_date	Date	No		
Reorder Level	reorder_level	Number	No		Used by DS-11 and DS-12
Batch Number	batch_number	Single Line	No		
Expiry Date	expiry_date	Date	No		
Location Code	location_code	Single Line	No		

Table 4.16: Field Specification – Product Orders Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Order Id	order_id	Auto Number	—	System generated, starts at 1	
Farmer	farmer	Lookup	No	Values from Farmers; sorted ascending; “Add New” enabled	
Bio Product	bio_product	Lookup	No	Values from Bio Products; sorted ascending; “Add New” enabled	Approval checked by DS-08
Order Quantity	order_quantity	Number	No		Validated by DS-08; stock by DS-09

Total Amount	total_amount	Currency (INR)	No		Computed by DS-08
Order Date	order_date	Date	No		

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Order Status	order_status	Drop Down	No	Delivered / Shipped / Confirmed / Cancelled / Pending / Processing (Others allowed)	
Delivery Address	delivery_address	Address	No		
Payment Status	payment_status	Drop Down	No	Partial / Paid / Unpaid (Others allowed)	
Notes	notes	Multi Line	No		

Table 4.17: Field Specification – Product Quality Tests Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Processing Batch	processing_batch	Lookup	No	Values from Processing Batches; sorted ascending; “Add New” enabled	
Bio Product	bio_product	Lookup	No	Values from Bio Products; sorted ascending; “Add New” enabled	
Test Date	test_date	Date	No		
Test Parameters	test_parameters	Multi Line	No		
Test Result	test_result	Drop Down	No	Pass / Review / Fail (Others allowed)	Defaulted by DS-04
Approval Status	approval_status	Drop Down	No	Approved / Rejected / Pending (Others allowed)	Triggers DS-03 and DS-05
Lab Name	lab_name	Single Line	No		
Technician Name	technician_name	Name	No		
Test Report Url	test_report_url	URL	No		

Table 4.18: Field Specification – Waste Categories Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Category Id	category_id	Auto Number	—	System generated, starts at 1	
Category Name	category_name	Single Line	No		
Description	description	Multi Line	No		
Unit Of Measurement	unit_of_measurement	Drop Down	No	Kilogram / Bale / Tonne / Cubic Meter (Others allowed)	
Recyclable Type	recyclable_type	Drop Down	No	Mixed / Organic / Inorganic (Others allowed)	

Hazardous Level	hazardous_level	Drop Down	No	High / Low / Medium (Others allowed)	
Storage Requirement	storage_requirement	Single Line	No		
Disposal Regulation	disposal_regulation	Single Line	No		

Table 4.19: Field Specification – Waste Collection Requests Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Request Id	request_id	Auto Number	—	System generated, starts at 1	
Waste Record	waste_record	Lookup	No	Values from Agricultural Waste Records; sorted ascending; "Add New" enabled	Validated by DS-02
Requested Quantity	requested_quantity	Number	No		Validated by DS-02
Collection Address	collection_address	Address	No		
Request Date	request_date	Date	No		
Request Status	request_status	Drop Down	No	Approved / Completed / Rejected / Pending (Others allowed)	Defaulted by DS-07; DS-06; Blueprint
Priority	priority	Drop Down	No	High / Low / Medium (Others allowed)	
Special Instructions	special_instructions	Multi Line	No		
Customer Contact	customer_contact	Phone	No		
Estimated Volume	estimated_volume	Number	No		

4.4 Process and Workflow Design

Figure 4.3 shows the end-to-end process and marks where Deluge scripts act (script identifiers are those of Table 5.2). Two sub-processes are controlled by Blueprint: the processing batch (Figure 4.4) and the waste collection request (Figure 4.5). In a Blueprint, a record can move only along the drawn transitions; the figures were produced from the transitions defined in the application.

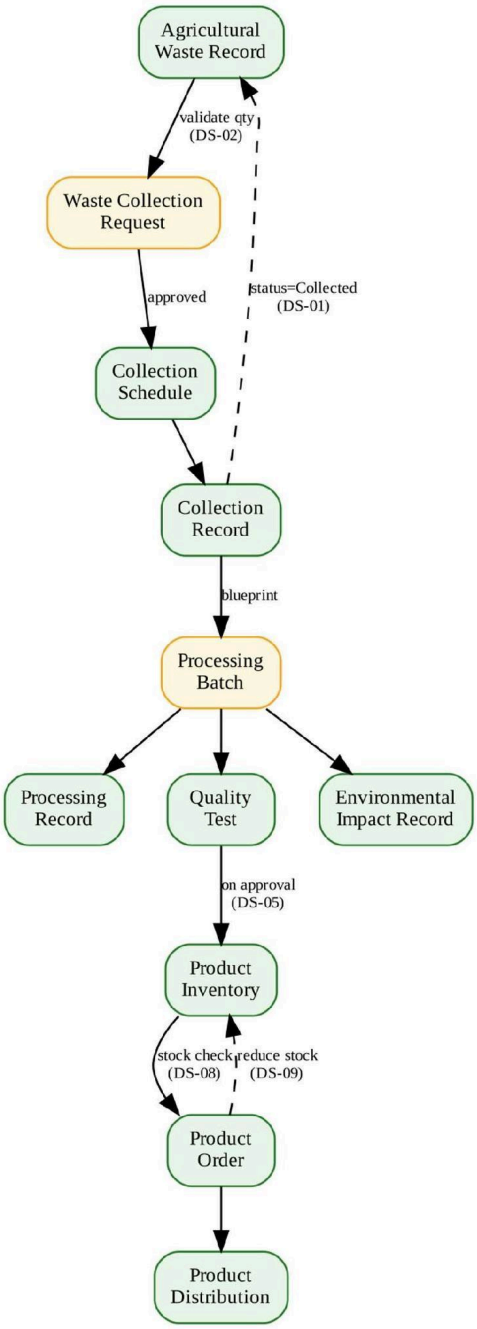


Figure 4.3: Workflow / Process Flow for the Waste-to-Product Chain

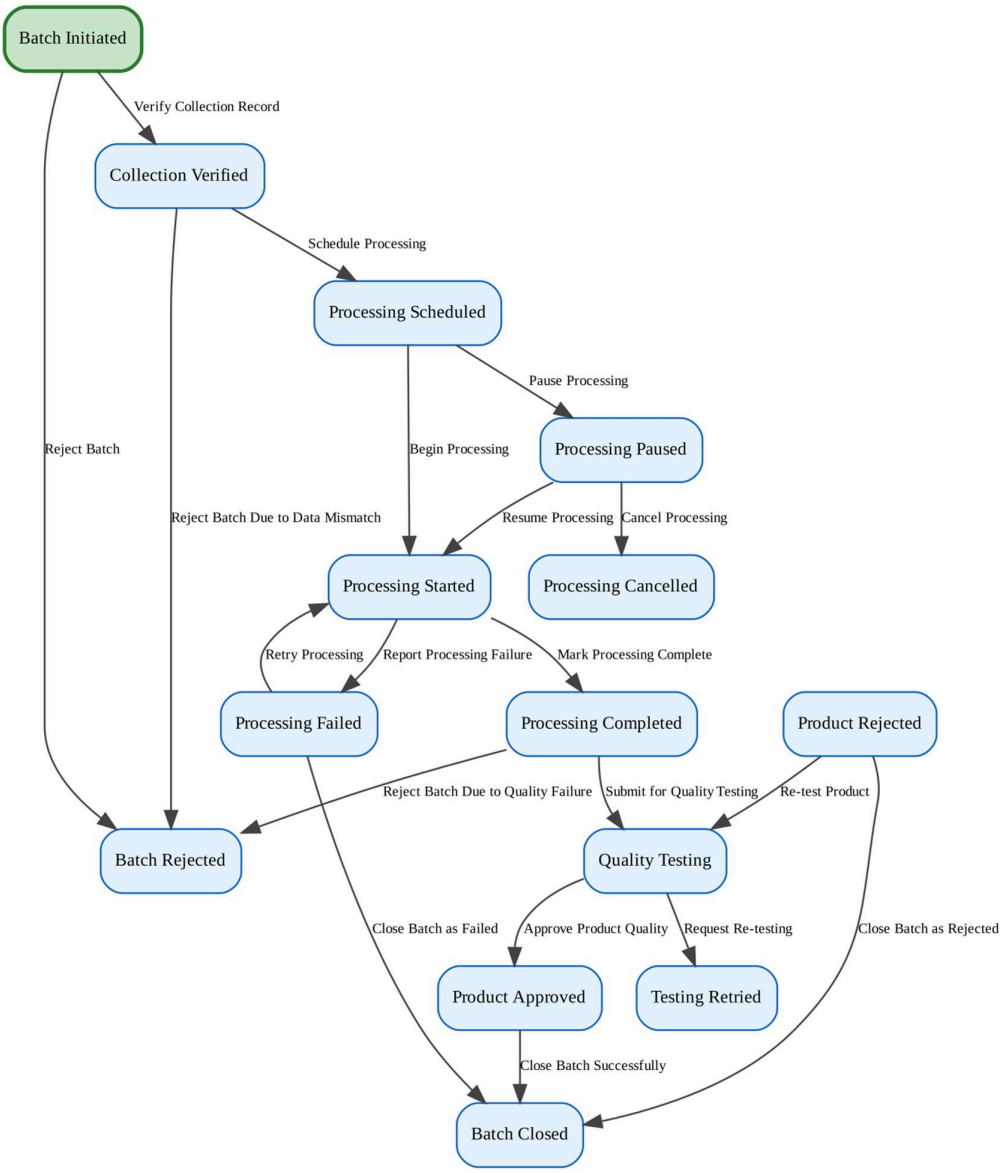


Figure 4.4: Blueprint Design – Bio Product Processing Workflow (Processing Batches)

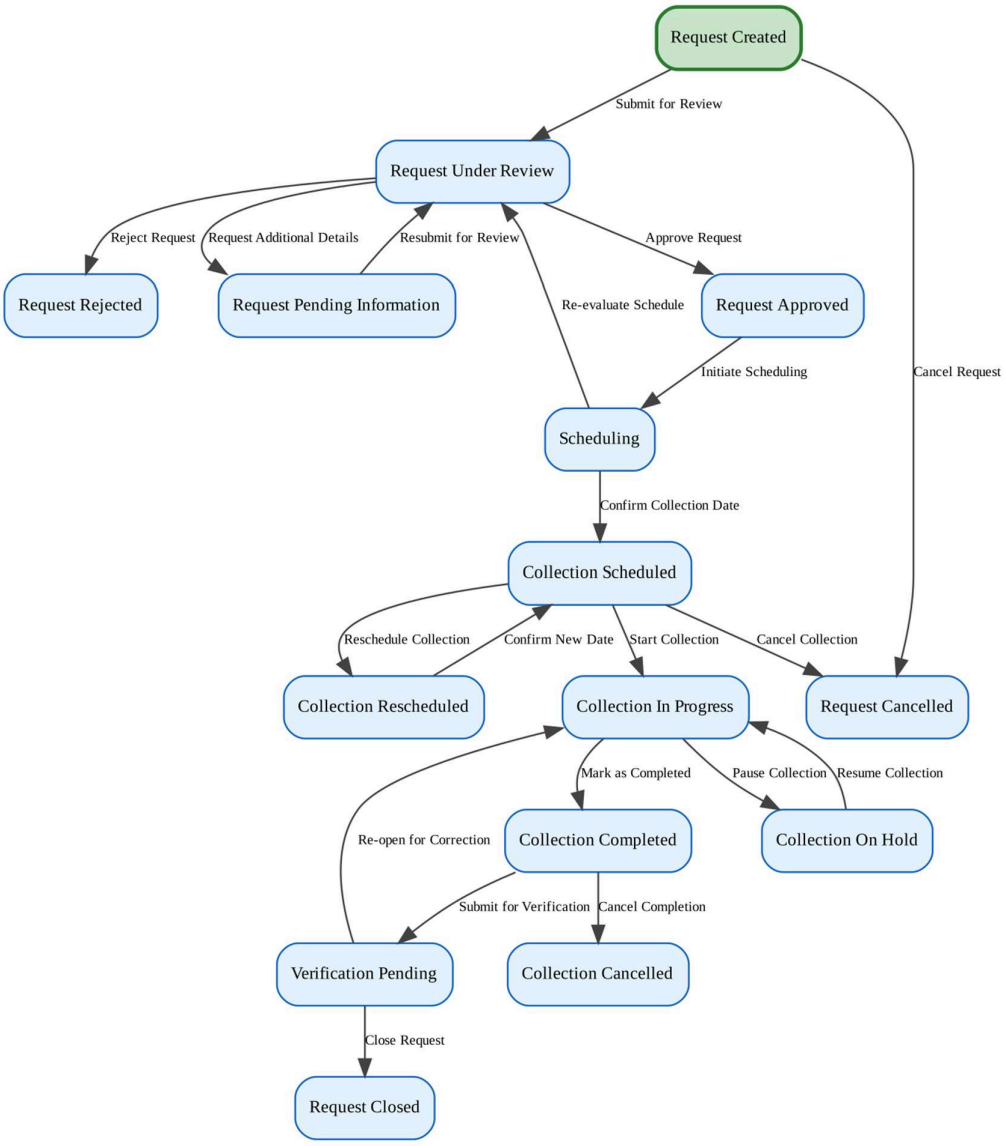


Figure 4.5: Blueprint Design – Waste Collection Request Workflow

4.5 User Interface and Navigation Design

Navigation follows the process order: master data first, then collection, processing, quality and inventory, and finally orders, distribution and impact. Each form has a list report and a kanban report in the menu. Users with the Administrator profile also see the Dashboard page, whose wireframe is shown in Figure 4.6: a row of four KPI tiles, two scatter charts, a status-count panel and an embedded list report.

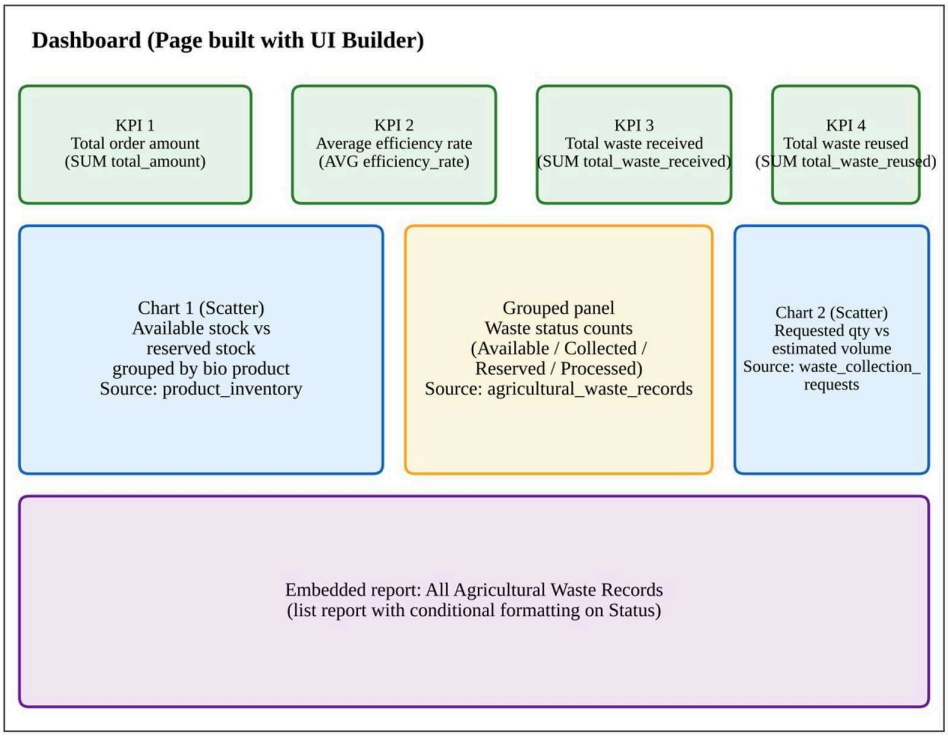


Figure 4.6: Navigation Structure and Wireframe of the Dashboard Page

4.6 Access Control Design

Seven custom permission profiles and one role were configured. “A E V D” means Add, Edit, View and Delete (Viewall and Modifyall in the platform); “V” means view only; “—” means no access. The Administrator profile has all permissions. Abbreviations: WCC – Waste Collection Coordinator; PCM – Processing Center Manager; ECO – Environmental Compliance Officer; IDS – Inventory and Distribution Specialist; Farmer and Buyer are customer-portal profiles.

Table 4.20: Role–Permission Matrix

Component (Form and its reports)	Admin	WCC	PCM	ECO	IDS	Farmer	Buyer
Form: Agricultural Waste Records	A E V D	—	—	—	—	—	—
Form: Bio Products	A E V D	—	—	—	A E V D	—	V
Form: Collection Records	A E V D	A E V D	—	—	—	—	—
Form: Collection Schedules	A E V D	A E V D	—	—	—	—	—
Form: Collection Teams	A E V D	A E V D	—	—	—	—	—
Form: Environmental Impact Records	A E V D	—	—	A E V D	—	—	—
Form: Farmers	A E V D	—	—	—	—	A E V D	—
Form: Farms	A E V D	—	—	—	—	A E V D	—
Form: Processing Batches	A E V D	—	A E V D	—	—	—	—
Form: Processing Centers	A E V D	—	A E V D	—	—	—	—
Form: Processing Methods	A E V D	—	A E V D	—	—	—	—
Form: Processing Records	A E V D	—	A E V D	—	—	—	—

Component (Form and its reports)	Admin	WCC	PCM	ECO	IDS	Farmer	Buyer
Form: Product Distribution	A E V D	—	—	—	A E V D	—	—
Form: Product Inventory	A E V D	—	—	—	A E V D	—	—
Form: Product Orders	A E V D	—	—	—	A E V D	A E V D	A E V D
Form: Product Quality Tests	A E V D	—	A E V D	A E V D	—	—	—
Form: Waste Categories	A E V D	—	—	V	—	—	—
Form: Waste Collection Requests	A E V D	A E V D	—	—	—	A E V D	—

CHAPTER 5 IMPLEMENTATION

5.1 Application Setup

The application “GreenAgro BioLoop” was created in the Zoho Creator workspace [from scratch / from a template / with the AI application builder] and was exported as a .ds definition (version 1.0, generated 07-Nov-2026). Application-level settings are date format dd-MMM-yyyy, time format 24-hour and time zone Asia/Kolkata, so that all dates and schedules follow Indian Standard Time. The application contains 18 forms, 33 reports, one page (Dashboard), 12 workflows (11 form workflows and one schedule) and two Blueprints. The menu is organised as [describe the sections you created, e.g. Master Data, Collection, Processing, Quality & Inventory, Sales & Distribution, Dashboard], and the theme/branding is [describe colours and logo].

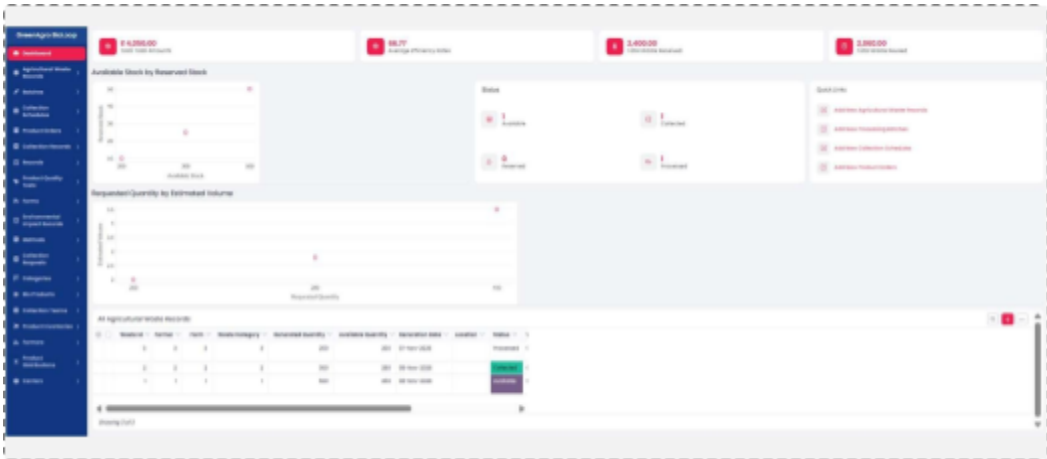


Figure 5.1: Application Home and Navigation Sections

5.2 Forms and Data Capture

The 18 forms contain 173 fields of 16 distinct types (Table 5.2). Master-data forms are listed first in Table 5.1, followed by the transactional forms in process order. Lookup fields (21 in all) are set to sort values in ascending order and to allow “Add New” so that a missing master record can be created without leaving the form. Choice fields have fixed options with an “Others” entry where the business may need extra values. The address field of the Agricultural Waste Records form captures coordinates and shows a map link, which the collection team can use for navigation.

Table 5.1: Forms Implemented

Module	Form	Fields	Lookups	Purpose
Master data	Farmers	9	0	Farmer profile, contact, primary crop
	Farms	10	1	Farm area, crop, soil and irrigation
	Waste Categories	8	0	Waste type, hazard level, storage and disposal rules
	Processing Centers	9	0	Centre capacity, manager, certification
	Processing Methods	8	0	Composting, vermicomposting, digestion parameters
	Collection Teams	10	0	Team leader, vehicle, capacity, district
	Bio Products	10	0	Product type, unit, selling price, shelf life
Collection	Agricultural Waste Records	11	3	Waste generated and available per farm
	Waste Collection Requests	10	1	Request to collect waste with priority
	Collection Schedules	10	2	Team, date, time and route of a pickup
	Collection Records	10	2	Quantity actually collected and quality check
Processing & quality	Processing Batches	10	3	Batch run at a centre with a method
	Processing Records	10	1	Output, residue, efficiency and grade
	Product Quality Tests	9	2	Laboratory tests and approval
	Environmental Impact Records	10	1	CO_2, water and land saved, reuse percentage
Sales & stock	Product Inventory	9	2	Stock by product and centre, reorder level
	Product Orders	10	2	Customer order with computed amount
	Product Distribution	10	1	Dispatch and delivery tracking

Table 5.2: Field Types Used Across the Forms

Field Type	Count	Example
Single Line	29	Agricultural Waste Records → Weather Condition
Number	27	Agricultural Waste Records → Generated Quantity
Drop Down	24	Agricultural Waste Records → Status
Lookup	21	Agricultural Waste Records → Farmer
Date	18	Agricultural Waste Records → Generation Date
Auto Number	13	Agricultural Waste Records → Waste Id
Multi Line	10	Agricultural Waste Records → Remarks
Name	8	Collection Schedules → Driver Name
Address	7	Agricultural Waste Records → Location

Phone	5	Collection Teams → Phone Number
Currency (INR)	2	Bio Products → Selling Price
Percent	2	Environmental Impact Records → Waste Reuse Percentage
Radio	2	Environmental Impact Records → Verification Status

Field Type	Count	Example
URL	2	Product Distribution → Proof Of Delivery
Time	1	Collection Schedules → Scheduled Time
Email	1	Farmers → Email
Multi Select	1	Processing Methods → Suitable Waste Types

To test the forms and reports with realistic values, a workbook of sample data (18 sheets, three records each, 54 records in all) was prepared for the Tamil Nadu setting: farmers in Thiruvallur, Pollachi and Lalgudi; paddy-straw, coconut-husk and banana waste; Chennai, Coimbatore and Trichy centres; and three products (organic compost, bio-fertiliser and biogas). Table 5.3 summarises the data set.

Table 5.3: Sample Data Prepared for the Application

Form / Sheet	Records	Illustrative Content
Farmers	3	Ramesh (Paddy), Lakshmi (Coconut), Murugan (Banana)
Farms	3	Ramesh Farm 5.5, Lakshmi Coconut Farm 3.2, Murugan Banana Farm 2.8 (area)
Waste Categories	3	Paddy Straw, Coconut Husk, Banana Waste (low hazard)
Processing Centers	3	Chennai, Coimbatore and Trichy Bio Centers; capacity 2000 / 1500 / 1200
Processing Methods	3	Composting (48 h, solar), Vermicomposting (72 h), Anaerobic Digestion (36 h)
Collection Teams	3	Team Alpha, Beta, Gamma with mini truck, pickup van, tractor trailer
Bio Products	3	Organic Compost ₹12, Bio Fertilizer ₹45, Biogas ₹30
Agricultural Waste Records	3	500 / 300 / 250 generated; statuses Available, Collected, Processed
Waste Collection Requests	3	Requests of 450 / 280 / 200; Approved, Pending, Approved
Collection Schedules	3	Pickups on 08-Nov and 09-Nov-2026 by Ravi, Kumar, Suresh
Collection Records	3	Collected 450 / 280 / 200; actual weights 448 / 278 / 198
Processing Batches	3	Input 448 / 278 / 198; In Progress, In Progress, Completed
Processing Records	3	Efficiency 71.4 / 68.3 / 60.6 %; grades A, A, B
Environmental Impact Records	3	Reuse 85 / 87.5 / 85 %; CO_2 saved 420 / 350 / 290
Product Orders	3	Orders of ₹1,200 (100 × 12), ₹2,250 (50 × 45), ₹600 (20 × 30)
Product Inventory	3	Stock 500 / 300 / 200 with reorder levels 100 / 75 / 50
Product Quality Tests	3	Two Approved, one Pending

Product Distribution	3	Dispatched, Delivered, In Transit

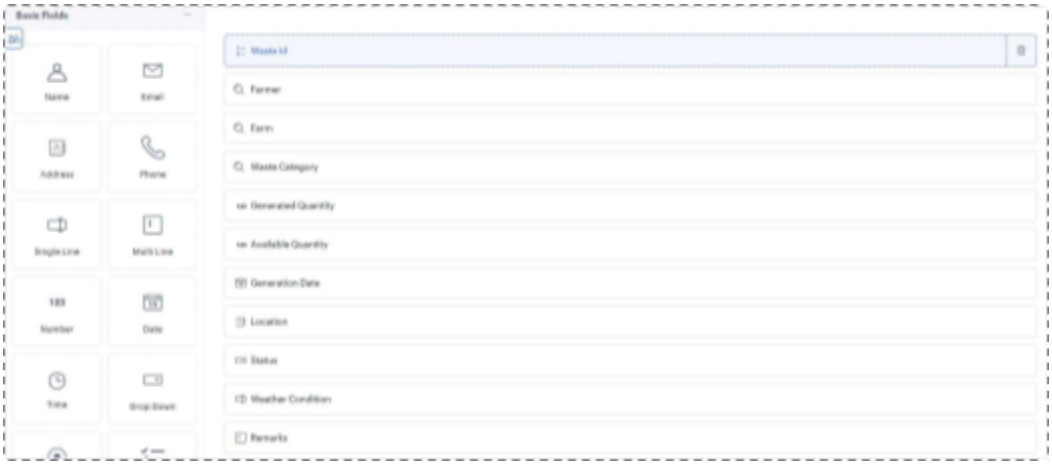


Figure 5.2: Agricultural Waste Records Form in the Form Builder



Figure 5.3: Agricultural Waste Records Form in Live Mode

5.3 Reports

A list report and a kanban report were created for each form where a status-type field exists, giving 33 reports: 17 list reports, 15 kanban reports and one custom report (All Farmers). List reports show all rows with conditional formatting on the status field; for example, in All Agricultural Waste Records the Status cell is purple for Available, green for Collected and red for Reserved. Kanban reports group cards by a status or classification field, such as Batch Status for processing batches or Delivery Status for distribution, so that a manager can see the work in each stage at a glance. Every report has a quick-view layout that shows all fields of a record. Access to each report follows the form permission of the profile (Table 5.4). Abbreviations are those of Section 4.6.

Table 5.4: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Agricultural Waste Records	List	Agricultural Waste Records	All rows; conditional formatting on status (3 rules)	None	Admin
Statuses	Kanban	Agricultural Waste Records	Cards grouped by status	None	Admin
All Batches	List	Processing Batches	All rows; conditional formatting on batch status (2 rules)	None	Admin, PCM
Batch Statuses	Kanban	Processing Batches	Cards grouped by batch status	None	Admin, PCM
All Collection Schedules	List	Collection Schedules	All rows; conditional formatting on status (3 rules)	None	Admin, WCC
Statuses	Kanban	Collection Schedules	Cards grouped by status	None	Admin, WCC
All Product Orders	List	Product Orders	All rows; conditional formatting on payment status (2 rules)	None	Admin, IDS, Farmer, Buyer
Order Statuses	Kanban	Product Orders	Cards grouped by order status	None	Admin, IDS, Farmer, Buyer
All Collection Records	List	Collection Records	All rows; conditional formatting on collection status (3 rules)	None	Admin, WCC
Collection Statuses	Kanban	Collection Records	Cards grouped by collection status	None	Admin, WCC
All Records	List	Processing Records	All rows; conditional formatting on processing status (3 rules)	None	Admin, PCM
Statuses	Kanban	Processing Records	Cards grouped by processing status	None	Admin, PCM
All Product Quality Tests	List	Product Quality Tests	All rows; conditional formatting on test result (3 rules)	None	Admin, PCM, ECO
Test Results	Kanban	Product Quality Tests	Cards grouped by test result	None	Admin, PCM, ECO
All Farms	List	Farms	All rows; conditional formatting on soil quality (2 rules)	None	Admin, Farmer
Soil Quality	Kanban	Farms	Cards grouped by soil quality	None	Admin, Farmer
All Environmental Impact Records	List	Environmental Impact Records	All rows	None	Admin, ECO
All Methods	List	Processing Methods	All rows; conditional formatting on energy source (4 rules)	None	Admin, PCM
Energy Sources	Kanban	Processing Methods	Cards grouped by energy source	None	Admin, PCM
All Collection Requests	List	Waste Collection Requests	All rows; conditional formatting on request status (4 rules)	None	Admin, WCC, Farmer
Request Statuses	Kanban	Waste Collection Requests	Cards grouped by request status	None	Admin, WCC, Farmer
All Categories	List	Waste Categories	All rows; conditional formatting on unit of measurement (1 rules)	None	Admin, ECO
Categories Board	Kanban	Waste Categories	Cards grouped by unit of measurement	None	Admin, ECO
All Bio Products	List	Bio Products	All rows; conditional formatting on product type (2 rules)	None	Admin, IDS, Buyer
Product Types	Kanban	Bio Products	Cards grouped by product type	None	Admin, IDS, Buyer

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
All Collection Teams	List	Collection Teams	All rows; conditional formatting on team availability (2 rules)	None	Admin, WCC
Team Availabilities	Kanban	Collection Teams	Cards grouped by team availability	None	Admin, WCC
All Product Inventories	List	Product Inventory	All rows	None	Admin, IDS
All Farmers	Custom	Farmers	All rows	None	Admin, Farmer
All Product Distributions	List	Product Distribution	All rows; conditional formatting on delivery status (3 rules)	None	Admin, IDS
Delivery Statuses	Kanban	Product Distribution	Cards grouped by delivery status	None	Admin, IDS
All Centers	List	Processing Centers	All rows; conditional formatting on certification level (2 rules)	None	Admin, PCM
Certification Levels	Kanban	Processing Centers	Cards grouped by certification level	None	Admin, PCM



Figure 5.4: All Agricultural Waste Records Report



Figure 5.5: Processing Batches Kanban Report

5.4 Pages and Dashboards

The Dashboard page was built with the UI Builder for the Administrator. Its top row has four KPI panels whose values are computed directly from form data: total order amount (sum of Total Amount in Product Orders), average efficiency rate (average of Efficiency Rate in Processing Records), total waste received and total waste reused (sums in Environmental Impact Records). Below are two scatter charts: Available Stock against Reserved Stock grouped by bio-product (source: Product Inventory) and Requested Quantity against Estimated Volume (source: Waste Collection Requests); a grouped panel with four counts of waste records by status; and an embedded All Agricultural Waste Records list report. Because the KPIs are bound to form data, the dashboard updates whenever a record is added or changed.

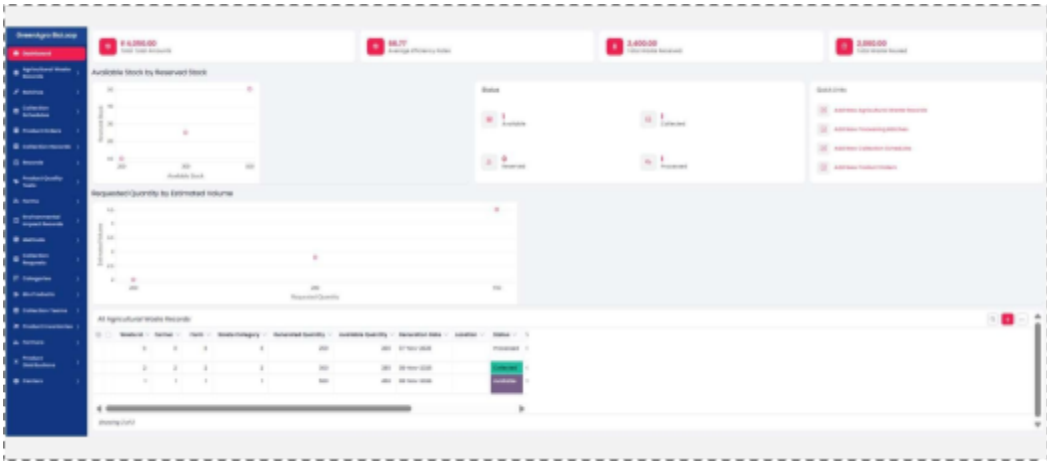


Figure 5.6: Dashboard Page

5.5 Deluge Scripting

Twelve Deluge scripts implement the business rules (Table 5.5). Eleven run as form workflows on form events (on validate, on success, on load, on update of a field) and one runs as a scheduled workflow. The common patterns are: fetching a related record by ID criteria (for example, the waste record behind a request), comparing values and stopping submission with “alert” and “cancel submit”, updating a field on a fetched record, inserting a record with “insert into”, and sending HTML e-mail with “sendmail”. The complete code is in Appendix C; three representative scripts are shown below.

Table 5.5: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Collection Records → On Add → On Success	Mark the linked waste record as Collected once a collection record is saved	fetch by ID, isNull(), field update
DS-02	Waste Collection Requests → On Add/Edit → On Validate	Block a request whose quantity exceeds the available waste quantity	fetch, if-else, alert, cancel submit
DS-03	Product Quality Tests → On Edit → On Update of Approval Status	Zero the saleable stock of a rejected product, append a rejection note and fail a Started batch	criteria fetch, for each, string concatenation, zoho.currentdate
DS-04	Product Quality Tests → On Add → On Load	Default the test result to Pending for a new test record	form load event, field default
DS-05	Product Quality Tests → On Edit → On Update of Approval Status	On approval, confirm an inventory row exists for the product and centre, creating one if missing	fetch, count(), insert into, info
DS-06	Waste Collection Requests → On Edit → On Update of Request Status	Log that an approved request is ready for collection	field-update event, info
DS-07	Waste Collection Requests → On Add → On Load	Default the request status to Pending for a new request	form load event, field default
DS-08	Product Orders → On Add/Edit → On Validate	Check product approval and stock, then compute the order total (quantity × selling price)	nested fetch, if-else, alert, arithmetic
DS-09	Product Orders → On Add → On Success	Reduce available stock by the ordered quantity after the order is saved	fetch, isNull(), arithmetic, field update, info
DS-10	Agricultural Waste Records → On Add/Edit → On Validate	Intended to check processing-centre capacity before accepting a waste record	alert, cancel submit
DS-11	Product Inventory → On Add/Edit → On Validate	Warn the user when available stock is below the reorder level	comparison, alert
DS-12	Schedule on Product Inventory (10:00 IST, Asia/Kolkata)	E-mail the administrator an HTML low-stock notice for each product below reorder level	schedule, sendmail, HTML message, string concatenation

Code 5.1: DS-02 – validation of the requested quantity

```
// DS-02 | Form: Waste Collection Requests | Event: On Validate
// Purpose: block a request that exceeds the available waste
// Fetch the waste record referenced in the request using the lookup
field
wasteRecord = agricultural_waste_records[ID == input.waste_record];
// Check if waste record exists
if(wasteRecord.isNull())
{
    alert "Error: Waste record not found for the selected waste.";
    cancel submit;
}
// Get the requested quantity from input
requestedQuantity = input.requested_quantity;
// Get the available quantity from the waste record
availableQuantity = wasteRecord.available_quantity;
// Validate if requested quantity exceeds available quantity
if(requestedQuantity > availableQuantity)
{
    alert "Error: Requested quantity (" +
requestedQuantity.toString() + ") exceeds available waste quantity
(" + availableQuantity.toString() + "). Please adjust your
request.";
    cancel submit;
```

```
}
```

Code 5.2: DS-09 – reduce stock after an order is saved

```
// DS-09 | Form: Product Orders | Event: On Add → On Success
// Purpose: reduce available stock by the ordered quantity
// Fetch the product order record that was just submitted orderRecord
= product_orders[ID == input.ID];
// Get the bio product and quantity from the order
selectedProduct = orderRecord.bio_product;
orderQuantity = orderRecord.order_quantity;
// Check if product and quantity are valid if(!selectedProduct.isNull()
&& !orderQuantity.isNull() && orderQuantity > 0)
{
    // Fetch the current inventory record for this product
    // Note: Adjust criteria if multiple inventory records exist per
product
    inventoryRecord = product_inventory[bio_product ==
selectedProduct];
    // If inventory record exists, reduce the available stock
    if(!inventoryRecord.isNull())
    {
        currentStock = inventoryRecord.available_stock; if(currentStock
        >= orderQuantity)
        {
            newStock = currentStock - orderQuantity;
            inventoryRecord.available_stock=newStock;
            info "Stock updated successfully. New available stock: "
+ newStock;
        }
        else
        {
            info "Error: Insufficient stock. Available: " +
currentStock + ", Required: " + orderQuantity;
        }
    }
    else
    {
        // If no inventory record exists, log error
        info "Warning: No inventory record found for product ID: "
+ selectedProduct;
    }
}
else
{
    info "Error: Invalid product or quantity in order.";
}
```

Code 5.3: DS-12 – scheduled low-stock e-mail (message text shortened)

```
// DS-12 | Schedule on Product Inventory | 10:00 IST
// Purpose: e-mail the administrator when stock < reorder level
if(input.available_stock < input.reorder_level)
{
    subject = "Low Stock Alert: " + input.bio_product.product_name;
    message = "<p>Dear Team,</p>" + "<p>The following product stock
has fallen below the reorder level:</p>" + ... ; sendmail
    [
        from :zoho.loginuserid
```

```
        to :zoho.adminuserid
        subject :subject
        message :message
        content type :HTML
    ]
}
```

DS-02 first fetches the waste record that the request points to, stops with an error if it cannot be found, and otherwise compares the requested quantity with the available quantity. When the request is too large it shows the two quantities in the message and cancels the submit, so that the record is not created. DS-09 runs only after a successful add: it fetches the order, finds the inventory row for the product, and subtracts the ordered quantity if enough stock exists, writing the new stock level to the log with "info". DS-12 loops over the inventory through the schedule and composes an HTML e-mail that names the product, available stock, reorder level, location code and last-updated date.

5.6 Workflows and Automation

The application has 11 form workflows, one schedule workflow and two Blueprints (Table 5.6). No approval process, batch workflow or report workflow was needed: approval of requests and of products is handled by status fields and the Blueprint stages. The schedule workflow "Monthly Inventory Reorder Alert" is currently saved with status inactive and must be activated before deployment (Section 7.1).

Table 5.6: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Auto-Set Waste Status on Collection	Form Workflow	Collection Records → on add; on success	—	Mark the linked waste record as Collected once a collection record is saved	DS-01
Validate Collection Request Quantity	Form Workflow	Waste Collection Requests → on add or edit; on validate	Requested qty > available qty	Block a request whose quantity exceeds the available waste quantity	DS-02
Exclude Product on Rejection	Form Workflow	Product Quality Tests → on edit; on update of approval_status	—	Zero the saleable stock of a rejected product, append a rejection note and fail a Started batch	DS-03
Set Pending Status on New Quality Test	Form Workflow	Product Quality Tests → on add; on load	—	Default the test result to Pending for a new test record	DS-04
Make Product Available on Approval	Form Workflow	Product Quality Tests → on edit; on update of approval_status	—	On approval, confirm an inventory row exists for the product and centre, creating one if missing	DS-05
Update Status on Approval	Form Workflow	Waste Collection Requests → on edit; on update of request_status	—	Log that an approved request is ready for collection	DS-06
Set Pending Status on New Request	Form Workflow	Waste Collection Requests → on add; on load	—	Default the request status to Pending for a new request	DS-07

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Validate Product and Calculate Sale Amount	Form Workflow	Product Orders → on add or edit; on validate	—	Check product approval and stock, then compute the order total (quantity × selling price)	DS-08
Reduce Stock on Sale Confirmation	Form Workflow	Product Orders → on add; on success	—	Reduce available stock by the ordered quantity after the order is saved	DS-09
Validate Storage Capacity on Create	Form Workflow	Agricultural Waste Records → on add or edit; on validate	—	Intended to check processing-centre capacity before accepting a waste record	DS-10
Show Low Stock Alert	Form Workflow	Product Inventory → on add or edit; on validate	—	Warn the user when available stock is below the reorder level	DS-11
Monthly Inventory Reorder Alert	Schedule	Daily/periodic at 10:00 IST from Last Updated Date	Available stock < reorder level	E-mail the administrator an HTML low-stock notice for each product below reorder level	DS-12
Bio Product Processing Workflow	Blueprint	Stage transitions on Processing Batches	—	14 stages, 19 transitions; starts at Batch Initiated	—
Waste Collection Request Workflow	Blueprint	Stage transitions on Waste Collection Requests	—	15 stages, 20 transitions; starts at Request Created	—

The Blueprint for processing batches begins at “Batch Initiated” and moves through verification, scheduling, processing, quality testing and approval to “Batch Closed”, with paths for rejection, pause, failure, cancellation and retesting (Table 5.7). The Blueprint for collection requests begins at “Request Created” and runs through review, approval, scheduling, collection and verification to “Request Closed”, with paths for information requests, rescheduling, hold, rejection and cancellation (Table 5.8). All transitions are of the normal type; no before/after actions or mandatory fields are configured on them yet, which is noted as an enhancement in Section 9.2.

Table 5.7: Blueprint Transitions – Bio Product Processing Workflow

Transition	From Stage	To Stage
Approve Product Quality	Quality Testing	Product Approved
Cancel Processing	Processing Paused	Processing Cancelled
Close Batch Successfully	Product Approved	Batch Closed
Close Batch as Failed	Processing Failed	Batch Closed
Close Batch as Rejected	Product Rejected	Batch Closed
Mark Processing Complete	Processing Started	Processing Completed
Pause Processing	Processing Scheduled	Processing Paused
Reject Batch	Batch Initiated	Batch Rejected
Reject Batch Due to Data Mismatch	Collection Verified	Batch Rejected
Reject Batch Due to Quality Failure	Processing Completed	Batch Rejected
Report Processing Failure	Processing Started	Processing Failed

Request Re-testing	Quality Testing	Testing Retried
Resume Processing	Processing Paused	Processing Started
Re-test Product	Product Rejected	Quality Testing

Transition	From Stage	To Stage
Retry Processing	Processing Failed	Processing Started
Schedule Processing	Collection Verified	Processing Scheduled
Begin Processing	Processing Scheduled	Processing Started
Submit for Quality Testing	Processing Completed	Quality Testing
Verify Collection Record	Batch Initiated	Collection Verified

Table 5.8: Blueprint Transitions – Waste Collection Request Workflow

Transition	From Stage	To Stage
Approve Request	Request Under Review	Request Approved
Cancel Collection	Collection Scheduled	Request Cancelled
Cancel Completion	Collection Completed	Collection Cancelled
Cancel Request	Request Created	Request Cancelled
Close Request	Verification Pending	Request Closed
Confirm Collection Date	Scheduling	Collection Scheduled
Confirm New Date	Collection Rescheduled	Collection Scheduled
Initiate Scheduling	Request Approved	Scheduling
Mark as Completed	Collection In Progress	Collection Completed
Pause Collection	Collection In Progress	Collection On Hold
Re-evaluate Schedule	Scheduling	Request Under Review
Reject Request	Request Under Review	Request Rejected
Re-open for Correction	Verification Pending	Collection In Progress
Request Additional Details	Request Under Review	Request Pending Information
Reschedule Collection	Collection Scheduled	Collection Rescheduled
Resubmit for Review	Request Pending Information	Request Under Review
Resume Collection	Collection On Hold	Collection In Progress
Start Collection	Collection Scheduled	Collection In Progress
Submit for Review	Request Created	Request Under Review
Submit for Verification	Collection Completed	Verification Pending

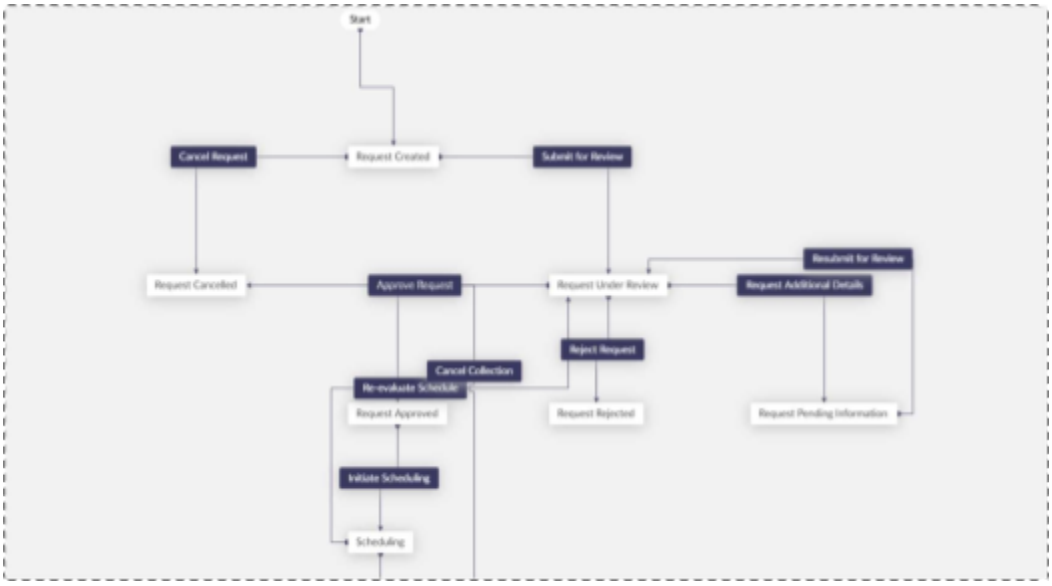


Figure 5.7: Workflow / Blueprint Configuration

5.7 Integrations

The integration needs of this version are modest, and the following are implemented (Table 5.9). The e-mail alert uses the Deluge sendmail task, which sends from the logged-in user's address to the administrator through Zoho Mail; no credentials are stored in the script. Address fields use the built-in "Link to Map" option so that collectors can open a location in a map. No external REST API, webhook, OAuth connection or Zoho CRM/Analytics integration was required because all data originates inside the application; the invokeurl task, a connection and the Creator REST API are the planned route for weather data, payment and accounting (Section 9.2).

Table 5.9: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Low-stock e-mail	Zoho Mail	Deluge sendmail task in a scheduled workflow (DS-12)	Logged-in Zoho account	HTML alert listing product, stock, reorder level, location code
Map link	Map service via Address field	Address field with "Link_to_Map" and coordinates	Not applicable	Opens collection and farm locations in a map
External REST API	None in this version	invokeurl GET/POST (planned)	API key / OAuth (planned)	Weather condition and alert fields are currently entered manually

5.8 User Management and Role-Based Access

Access is implemented through permission profiles (Table 5.10). Four are internal operational profiles with module permissions limited to the forms of their function;

“Administrator” holds all permissions; “Read” and “Write” are general-purpose profiles; “Developer” is the developer profile. Two customer-portal profiles, Farmer and Bio-Product Buyer, plus the default Customer profile, let external parties use the application without an internal licence. One role, CEO, is defined, whose members can access the data of all other users. The matrix of Section 4.6 was implemented as View-all and Modify-all permissions per form, which also grant access to the form’s reports.

Table 5.10: Permission Profiles Configured

Profile	Type	Description	Forms Accessible
Read	Users	This profile will have read permission for all components	Read access to all components
Write	Users	This profile will have write permission for all components	Write access to all components
Inventory and Distribution Specialist	Users	Manage stock levels of bio-products across different processing centers	Product Distribution, Product Orders, Bio Products, Product Inventory
Administrator	Users	This profile will have all the permissions.	All components
Environmental Compliance Officer	Users	Calculate and record environmental impact metrics like CO2 and water savings	Product Quality Tests, Environmental Impact Records, Waste Categories
Waste Collection Coordinator	Users	Review and approve waste collection requests based on priority and availability	Waste Collection Requests, Collection Records, Collection Schedules, Collection Teams
Processing Center Manager	Users	Oversee daily operations at processing centers and manage batch schedules	Processing Records, Product Quality Tests, Processing Batches, Processing Methods, Processing Centers
Developer	Developer	Developer Profile	—
Farmer	Customer Portal	Register farm details and report generated agricultural waste quantities	Product Orders, Waste Collection Requests, Farmers, Farms
Bio-Product Buyer	Customer Portal	View product specifications and quality grades before purchasing	Bio Products, Product Orders
Customer	Customer Portal	This is the default profile having only add and view permission.	—

Two points need attention before go-live. First, the Farmer portal profile can view and modify all records of the forms it can reach, because no record-level criteria (for example, “Farmer == logged-in user”) are configured; a farmer could therefore see other farmers’ orders. Second, the Farmer profile has no access to the Agricultural Waste Records form even though its purpose is to report generated waste. Both are logged in Table 6.2.

5.9 AI and Analytics Features

Analytics. The dashboard (Section 5.4) provides KPI figures computed from live data (sum, average and count aggregations) and two scatter charts; the kanban reports act as visual analytics over status fields. A Zoho Analytics integration was not used.

AI. [State clearly whether Zia or the AI application builder was used, for example: “The Deluge scripts were first drafted with the AI assistant in the Deluge editor and then reviewed, tested and corrected by me.” If AI assistance was used, name the scripts concerned and describe how each was verified.] The scripts in Appendix C contain explanatory comments of the kind produced by an assistant; any such content must be acknowledged here, as required by the declaration of this report.

5.10 Pro-Code Extension

No pro-code extension was used. All requirements were met with configuration and Deluge, which keeps the application maintainable by a business analyst. If volume grows, the planned extensions are: a Zoho Catalyst function to compute emission savings from batch data, a JavaScript widget for a map of collection routes, and Creator REST API calls to feed a Zoho Analytics workspace for advanced dashboards.

5.11 Mobile and Portal Access

Because Zoho Creator generates a mobile app for every application, the forms and reports are available on Android and iOS without extra development; collection teams can use the mobile app to open request details and map links in the field. The Farmer and Bio-Product Buyer profiles are customer-portal profiles, which allow external users to sign in to a portal view of the application.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was planned at four levels. **Unit testing** exercised each form on its own: a record was added with valid and invalid data and the success message, defaults and lookups were checked. **Script testing** exercised each Deluge script with inputs on both sides of its condition (for example, a requested quantity equal to and above the available quantity). **Integration testing** followed a record through the chain waste → request → schedule → collection → batch → quality test → inventory → order → distribution and checked that the status changes and stock updates of DS-01, DS-05 and DS-09 occurred. **Role-based testing** signed in as a user of each permission profile and confirmed that only the permitted forms and reports were visible. Finally, user acceptance testing was carried out with end users (Section 6.5). The 54 sample records described in Section 5.2 were used as the test data.

6.2 Test Cases

Table 6.1 lists 22 test cases covering every functional requirement of Table 3.2, including negative tests. The columns "Actual Result" and "Status" are to be completed from the execution of each test on the live application.

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Add a farmer with all details	Farmers form: Ramesh, +919876500001, Paddy	Record created; "Farmers Added Successfully"	[ ]	[ ]
TC-02	FR-01	Add a farm linked to a farmer	Farms form: choose farmer from lookup, area 5.5	Farm saved; farmer shown through the lookup	[ ]	[ ]
TC-03	FR-02	Add a waste record	Farmer 1, Farm 1, Paddy Straw, 500 / 450, status Available	Record saved with auto-number Waste Id	[ ]	[ ]
TC-04	FR-03	Raise a request within available quantity	Waste record 1, requested 450	Request saved; status Pending	[ ]	[ ]
TC-05	FR-03	Negative: request above available quantity	Waste record 1, requested 600	Error showing both quantities; record not saved	[ ]	[ ]
TC-06	FR-03	Default status on new request	Open the Waste Collection Requests add form	Request Status shows Pending (DS-07)	[ ]	[ ]
TC-07	FR-04	Request Blueprint transition	Request Under Review → Approve Request	Stage becomes Request Approved; only valid transitions offered	[ ]	[ ]

TC-08	FR-05	Auto-mark waste as collected	Add a collection record for waste record 1	Waste record status becomes Collected (DS-01)	[ ]	[ ]

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-09	FR-06	Create a processing batch	Collection record 1, centre 1, Composting, input 448	Batch saved; Blueprint at Batch Initiated	[ ]	[ ]
TC-10	FR-04	Batch Blueprint to approval	Quality Testing → Approve Product Quality	Stage becomes Product Approved	[ ]	[ ]
TC-11	FR-07	Default test result	Open the Product Quality Tests add form	Test Result shows Pending (DS-04)	[ ]	[ ]
TC-12	FR-07	Approve a quality test	Set Approval Status = Approved	Inventory row for product and centre exists or is created (DS-05)	[ ]	[ ]
TC-13	FR-07	Negative: reject a quality test	Set Approval Status = Rejected	Available stock of the product becomes 0; rejection note appended (DS-03)	[ ]	[ ]
TC-14	FR-09	Place a valid order	Farmer 1, Organic Compost, 100 (price 12)	Total ₹1,200; stock reduced by 100 (DS-08, DS-09)	[ ]	[ ]
TC-15	FR-09	Negative: order above stock	Organic Compost, quantity larger than stock	“Insufficient stock” message; order not saved	[ ]	[ ]
TC-16	FR-08	Low-stock warning	Inventory: available 40, reorder level 50	Low Stock Alert message (DS-11)	[ ]	[ ]
TC-17	FR-08	Scheduled e-mail	Activate and run DS-12 with stock below reorder level	E-mail received by the administrator	[ ]	[ ]
TC-18	FR-10	Record a dispatch	Distribution for order 1, status Dispatched	Record saved; visible in kanban by delivery status	[ ]	[ ]
TC-19	FR-11	Record environmental impact	Batch 1: received 1000, reused 850, CO_2 420	Record saved; dashboard totals include it	[ ]	[ ]
TC-20	FR-12	Dashboard totals	Open Dashboard as Administrator	KPI values equal sums/averages in the forms	[ ]	[ ]
TC-21	FR-13	Kanban grouping	Open Processing Batches kanban	Cards grouped by Batch Status	[ ]	[ ]
TC-22	FR-14 / NFR-02	Negative: unauthorised access	Sign in as Farmer; look for Processing Batches	Form and its reports not listed	[ ]	[ ]

6.3 Debugging Deluge Scripts

Three techniques were used. The Deluge editor’s syntax check flagged unknown fields and missing semicolons before saving. “info” statements, already present in DS-05, DS-06 and DS-09, write the values of variables and the decision taken to the execution log, which was read after each test. Error messages shown through “alert” make validation failures visible to the user and were used to confirm which branch ran.

One real defect found by walking through the scripts was in DS-10 (Validate Storage Capacity on Create). The script was meant to refuse a waste record when a processing centre’s capacity is exceeded, but the Agricultural Waste Records form has no processing-centre field. The script therefore ends in an unconditional “alert” and “cancel

submit", which would block every add and edit of a waste record. The fix is to remove it from this form and place a real capacity check on the Processing Batches form, where the centre is selected (Code 6.1). This is recorded as BUG-01 in Table 6.2.

Code 6.1: Recommended replacement for DS-10 on the Processing Batches form

```
// DS-10 (revised) | Form: Processing Batches | Event: On Validate
// Purpose: refuse a batch larger than the centre's processing
capacity
center = processing_centers[ID == input.processing_center];
if(!center.isNull() && input.input_quantity >
center.processing_capacity)
{
    alert "Input quantity exceeds the capacity of " +
center.center_name;
    cancel submit;
}
```

Figure 6.1: Deluge Debug Output / Log

6.4 Defect Log

Table 6.2 records defects found during the review of the exported application and its sample data. All entries are open until the corrective action has been applied and re-tested; update the Status column accordingly.

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied / Recommended	Status
BUG-01	Every attempt to add or edit an Agricultural Waste Record is refused	DS-10 refers to a field that does not exist in the form and ends with an unconditional cancel submit	Delete DS-10 from this form; add a capacity check to Processing Batches (Code 6.1)	Open
BUG-02	Orders for unapproved or out-of-stock products are not stopped	DS-08 uses "return" after alert instead of "cancel submit"	Replace "return" with "cancel submit" in each alert branch	Open
BUG-03	Stock may be read from or reduced at the wrong centre	DS-08 and DS-09 take the first inventory row of a product, but a product can be stocked at several centres (sample data: Organic Compost at two centres)	Add a centre to the order or sum stock across rows and reduce by earliest expiry	Open
BUG-04	No low-stock e-mail is sent	DS-12 is saved with status inactive and is tied to the Last Updated Date	Activate the schedule and set a recurring monthly or daily frequency	Open
BUG-05	Rejecting one batch zeroes stock of the product at all centres	DS-03 loops over every inventory row of the product	Filter inventory by the processing centre of the rejected batch	Open
BUG-06	Approving a request does nothing visible	DS-06 only writes an info message	Add a notification to the team or create a collection schedule record; otherwise remove	Open
BUG-07	Sample data values do not match 15 picklist columns (e.g. Collection Status "Completed" vs "Collected", Quality Grade "A" vs "Grade A"); Suitable Waste Types values are not in the multi-select list	Workbook prepared independently of the form options; most fields allow "Others" but the multi-select does not	Normalise the workbook to the option values or add the options to the forms before import	Open
BUG-08	Test Result default "Pending" is not a listed option	DS-04 sets a value outside Pass / Review / Fail	Add "Pending" to the Test Result options	Open
BUG-09	Farmer portal user can see all farmers' orders and cannot reach waste records	No record-level criteria; Agricultural Waste Records not granted to the profile	Add criteria "farmer == logged-in user" and grant the form	Open

Bug ID	Description	Root Cause	Fix Applied / Recommended	Status
BUG-10	Incomplete records can be saved	No field is marked mandatory	Mark key fields (farmer, quantity, dates, lookups) as mandatory	Open

6.5 User Acceptance Testing

Feedback is to be recorded from at least two users who operated the application in their assigned roles.

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
[ ]	[ ]	[ ]	[ ]
[ ]	[ ]	[ ]	[ ]

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments. [Describe what was done, for example: “The application was built and tested in the Development environment, promoted to Stage for user acceptance testing with the sample data, and published to Production on DD Month YYYY with version notes. Before publishing, the sample records were removed and the production users and portal profiles were created. The previous published version is kept as the rollback point.”] Table 7.1 is the checklist used before publishing; two items specific to this application are the activation of the schedule workflow (BUG-04) and the closure of BUG-01 and BUG-02.

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	[ ]
2	Sample/test data removed or isolated	[ ]
3	Roles, permissions and users configured in Production	[ ]
4	Connections authorised in Production (none required in this version)	[ ]
5	Schedules and workflows enabled (DS-12 is currently inactive)	[ ]
6	Application published and URL verified	[ ]
7	Version / release notes recorded	[ ]
8	Defects BUG-01 and BUG-02 closed and re-tested	[ ]

7.2 Performance Optimisation

Performance was addressed in the way scripts and screens are written. Scripts fetch a single related record by its ID rather than scanning a form (DS-01, DS-02, DS-09), and the only loop (DS-03) iterates over rows already fetched with one criteria expression, so there is no fetch inside a loop. Dashboard figures are computed by the platform’s aggregation on the source form (sum, average, count) instead of by scripts. Table 7.2 is to be completed with measured values from the live application.

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Fetch by ID criteria instead of scanning the form	DS-01, DS-02, DS-09	[ ]	[ ]	[ ]

Optimisation Applied	Component	Before	After	Improvement
Single criteria fetch before the loop (no fetch inside the loop)	DS-03	[ ]	[ ]	[ ]
KPIs computed by aggregation on the source form	Dashboard	[ ]	[ ]	[ ]

7.3 Security Features

Security rests on permission profiles, separation of portal and internal users, validation scripts and the absence of stored credentials (Table 7.3). The last three rows are not yet configured and are recommended before go-live.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Role / profile permissions (implemented)	Seven custom profiles with View-all / Modify-all limited to the forms of each function	Staff reaching data outside their duties
Customer-portal separation (implemented)	Farmer and Bio-Product Buyer profiles of type Customer Portal	External users seeing internal screens
Server-side validation (implemented)	DS-02, DS-08, DS-11 on validate	Invalid quantities and orders
No hard-coded credentials (implemented)	Only sendmail through the logged-in Zoho account is used	Credential leakage
Record-level criteria (recommended)	Farmer and Buyer profiles on Product Orders and Requests	A portal user reading others' records
Mandatory fields (recommended)	Key fields of all forms	Incomplete or orphan records
Two-factor authentication and IP restriction (recommended)	Organisation settings	Account takeover

7.4 Governance and Administration

Naming follows one convention: form and field link names in lower-case with underscores, report names ending in "_Report" for list reports and "_by_<field>" for kanban reports, and workflow display names that state the action (for example, "Reduce Stock on Sale Confirmation"). The application definition is exported as a .ds file, which serves as a backup and as a record of version 1.0. [Describe the audit trail you reviewed, how user licences are managed, and who is responsible for maintenance.]

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working of the application is shown by following one batch of paddy straw from the sample data through each role. Farmer Ramesh of Thiruvallur has generated 500 units of paddy straw, of which 450 are available on 08-Nov-2026. (1) The Administrator or farmer records the waste in Agricultural Waste Records (Figure 8.1).

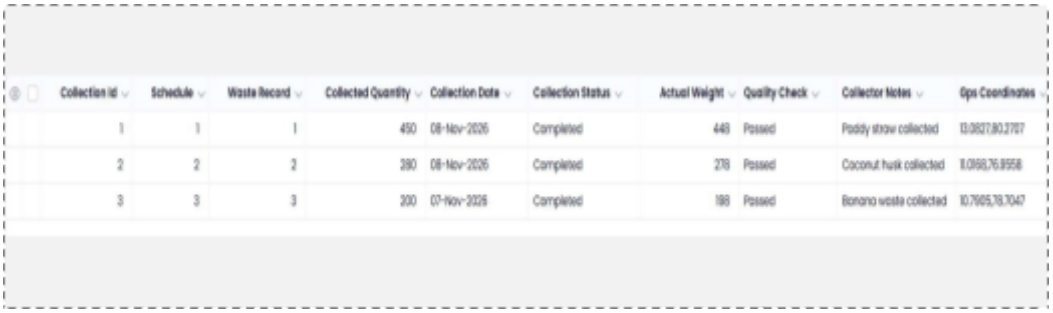
(2) A request for 450 is raised and, being within the available quantity, passes DS-02; it is approved by the Waste Collection Coordinator (Figure 8.2). (3) Team Alpha is scheduled at 09:00 and the collection is recorded as 450 collected and 448 actual; DS-01 marks the waste Collected (Figure 8.3). (4) The Processing Center Manager starts a composting batch of 448 at the Chennai centre, and the Blueprint carries it through its stages (Figure 8.4). (5) The processing record shows 320 output and 71.4 % efficiency, and a quality test for Organic Compost is Approved (Figure 8.5). (6) Ramesh orders 100 units at ₹12, the total ₹1,200 is computed by DS-08, and stock falls through DS-09; the dashboard totals then include the order (Figure 8.6).



Figure 8.1: Scenario Step – Waste record created for farmer Ramesh

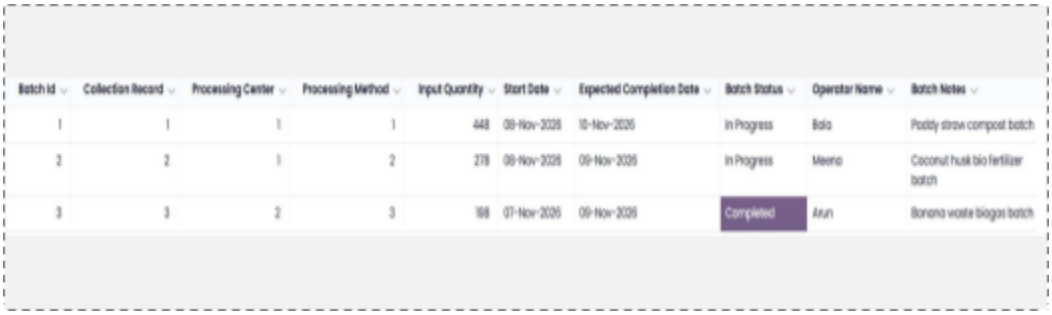


Figure 8.2: Scenario Step – Collection request validated and approved



Collection Id	Schedule	Waste Record	Collected Quantity	Collection Date	Collection Status	Actual Weight	Quality Check	Collector Notes	Gps Coordinates
1	1	1	450	08-Nov-2026	Completed	448	Passed	Paddy straw collected	13.0827,80.2707
2	2	2	280	08-Nov-2026	Completed	278	Passed	Coconut husk collected	11.0168,76.9558
3	3	3	200	07-Nov-2026	Completed	198	Passed	Banana waste collected	10.7905,78.7047

Figure 8.3: Scenario Step – Collection scheduled and recorded; waste marked Collected



Batch Id	Collection Record	Processing Center	Processing Method	Input Quantity	Start Date	Expected Completion Date	Batch Status	Operator Name	Batch Notes
1	1	1	1	448	08-Nov-2026	10-Nov-2026	In Progress	Bala	Paddy straw compost batch
2	2	1	2	278	08-Nov-2026	09-Nov-2026	In Progress	Meena	Coconut husk bio fertilizer batch
3	3	2	3	198	07-Nov-2026	09-Nov-2026	Completed	Arun	Banana waste biogas batch

Figure 8.4: Scenario Step – Processing batch moving through the Blueprint



Processing Batch	Bio Product	Test Date	Test Parameters	Test Result	Approval Status	Lab Name	Technician Name	Test Report Url
1	1	10-Nov-2026	Moisture, pH, NPK	Within limits	Approved	AgriLab Chennai	Kavitha	https://example.com/report
2	2	09-Nov-2026	pH, nutrient content	Within limits	Approved	AgriLab Coimbatore	Mohan	https://example.com/report2
3	3	09-Nov-2026	Gas purity, methane level	Passed	Pending	BioLab Trichy	Siva	https://example.com/report3

Figure 8.5: Scenario Step – Processing record and approved quality test

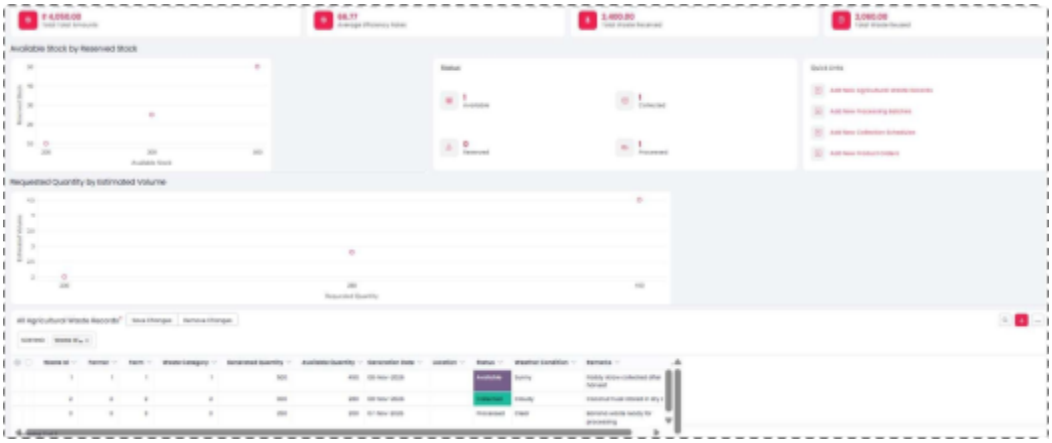


Figure 8.6: Scenario Step – Order placed; dashboard updated

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Normalised data model of 18 linked forms	Sections 4.2, 4.3, 5.2	18 forms, 173 fields, 21 lookup relationships (Tables 4.1 to 4.19, 5.1)
Automation with Deluge and workflows	Sections 5.5, 5.6	12 scripts: 11 form workflows and one schedule (Tables 5.5, 5.6)
Blueprints for batches and requests	Section 5.6	14-stage and 15-stage Blueprints (Tables 5.7, 5.8)
Reports and a management dashboard	Sections 5.3, 5.4	33 reports and one dashboard page (Table 5.4)
Role-based access including portal users	Sections 4.6, 5.8	Seven custom profiles and one role (Tables 4.20, 5.10)
Secure deployment with governance	Chapter 7	Checklist, security controls and naming conventions (Tables 7.1, 7.3); items marked "recommended" remain

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Time to process one request	Several phone calls and manual entries [measured time]	One form submission with validation [measured time]	Fewer hand-offs
Manual steps	Nine hand-offs between phone, register, slips and spreadsheets	Data entered once and reused by lookup; statuses and stock updated by workflows	Re-entry removed
Report generation	Month-end manual compilation	Instant list and kanban reports and dashboard KPIs	Real-time visibility
Data accessibility	Local sheets and paper	Web and mobile, by role; portal for farmers and buyers	Anywhere, controlled access
Traceability	None	Order → product → batch → collection → waste → farm	End-to-end

8.4 Limitations of the Low-Code Approach

Four constraints were met. First, **Deluge and platform limits**: scripts are bound by per-execution statement limits and plan-based API-call limits, so heavy batch logic is better placed in a function or an external service; this also influenced the choice to avoid loops over large forms. Second, **modelling limits**: a lookup picks one parent, so a product stocked at several centres (BUG-03) needs explicit design of the inventory key, and the same status information exists both in a field and in a Blueprint stage, which must be kept consistent by convention. Third, **UI customisation**: only scatter charts and KPI panels were used on the dashboard; more specialised visuals need the page

builder’s

scripting or a pro-code widget. Fourth, **vendor lock-in**: the application, its Deluge code and its data model live on one platform, so a move elsewhere would mean rebuilding [5]. Workarounds were to keep the logic in short, named scripts, to export the .ds definition regularly, and to keep the data model in a portable form (Figure 4.2).

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application is **no-code configuration**: the 18 forms, 33 reports, dashboard, profiles and Blueprints were built by dragging and setting properties. **Low-code** was needed for the 12 Deluge scripts that enforce rules such as quantity validation, status propagation and stock updates. **Pro-code** has not been needed yet; it would be justified if the organisation required real-time route optimisation, heavy computation of emission factors, or a custom mobile experience for drivers. The decision rule adopted is to stay on the platform while a requirement can be met by configuration or a short script, and to move to Catalyst or external code only when a script becomes long, slow or needs libraries the platform lacks.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

GreenAgro BioLoop brings the whole agricultural-waste-to-bio-product chain into one Zoho Creator application: 18 linked forms, 33 reports, a dashboard, 12 Deluge workflows, two Blueprints and seven permission profiles. It replaces phone calls, registers and disconnected spreadsheets by a single record that follows the material from the farm to the customer, with validations, status updates and stock control done by the system.

The work also showed the practical limits of the approach. The review of the application found defects (Table 6.2), chiefly in scripts that were not fully aligned with the data model, and showed that low-code does not remove the need for careful testing. With those corrected and the recommended security settings applied, the application is ready for pilot use by the organisation's operations teams.

9.2 Future Enhancements

- Fetch weather conditions and alerts for the collection date from a weather API through invokeurl and a connection, replacing manual entry.
- Add online payment and invoice generation for product orders, and sync customers and orders to an accounting or CRM application.
- Configure Blueprint before/after actions and mandatory fields so that, for example, a batch cannot move to "Quality Testing" without an output record.
- Add record-level criteria and field-level hiding for portal users, and two-factor authentication for internal users.

9.3 Learning Outcomes and Career Relevance

The project gave hands-on practice in data modelling with lookups, form and report design, Deluge scripting, workflow and Blueprint design, role-based security, testing and deployment on a low-code platform. These are the day-to-day skills of a low-code developer and of a solutions consultant or automation engineer. Reviewing one's own scripts for correctness, as done in Section 6.3, is also the habit that separates a citizen developer from a professional one.

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APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

"Y" means implemented and verifiable in the exported application; "Partial" means implemented in part (see the note); "N" means not implemented; a highlighted entry must be confirmed against the live application.

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Application "GreenAgro BioLoop"	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	18 forms; 16 field types	5.2	Tables 5.1, 5.2
II	Lookup field (relationship between forms)	Y	21 lookup fields	4.2, 5.2	Fig. 4.2
II	Subform (parent–child data)	N	Not used; children are separate forms	4.2, 5.2	—
II	Field validations / layout rules	Y	Deluge validation: DS-02, DS-08, DS-11	5.2	Code 5.1
II	List report with filter, grouping, custom action	Partial	17 list reports with conditional formatting; grouping, filters and custom actions [confirm]	5.3	Fig. 5.4
II	Kanban / Calendar / Timeline / Map report	Y	15 kanban reports	5.3	Fig. 5.5
II	Pivot table / chart report	N	No pivot report; charts are on the Dashboard page	5.3	—
II	Page / dashboard built with UI Builder	Y	Page "Dashboard"	5.4	Fig. 5.6
II	Deluge – form events (validate / success / field actions)	Y	DS-01 to DS-11	5.5	Code 5.1, 5.2
II	Deluge – functions, fetch, loops, collections	Partial	fetch, if-else and for each used; no custom functions	5.5	Appendix C
II	Email / in-app notifications	Partial	sendmail in DS-12 (schedule currently inactive)	5.5, 5.6	Code 5.3
II	Form workflow	Y	11 form workflows	5.6	Table 5.6
II	Scheduled workflow	Partial	Monthly Inventory Reorder Alert (inactive)	5.6	Table 5.6
II	Approval process	N	Approval handled by status fields and Blueprint	5.6	—
II	Blueprint	Y	Two Blueprints (14 and 15 stages)	5.6	Tables 5.7, 5.8
II	Batch / report workflow	N	Not used	5.6	—
II	Testing & debugging evidence	[Y / N]	[Test execution evidence]	6	Table 6.1
III	Integration with a Zoho service	Partial	Zoho Mail through sendmail	5.7	Table 5.9
III	Connection / REST API call (invokeurl)	N	Not used in this version	5.7	—
III	Webhook or Creator REST API	N	Not used in this version	5.7	—
III	Users, roles and permission profiles	Y	7 custom profiles; role CEO	5.8	Table 5.10
III	Record-level access criteria / customer portal	Partial	Portal profiles defined; record-level criteria not set	5.8	Table 4.20
III	Zia / AI feature	[Y / N]	[ ]	5.9	[ ]
III	Analytics (charts / Zoho Analytics)	Y	Dashboard KPIs and two charts	5.9	Fig. 5.6

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
IV	Performance optimisation applied & measured	[Y / N]	[ ]	7.2	Table 7.2
IV	Deployment across environments	[Y / N]	[ ]	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	Partial	Profile permissions; others recommended	7.3	Table 7.3
IV	Governance (audit trail, administration)	[Y / N]	[ ]	7.4	Fig. 7.2
V	Pro-code extension (Catalyst / widgets / custom functions)	N	Justified in Section 5.10	5.10	—
V	Low-code limitations & spectrum analysis	Y	Written analysis	8.4, 8.5	—
—	Mobile app / portal access	[Y / N]	Portal profiles defined; mobile app [confirm]	5.11	Fig. 5.10

Total features implemented: **12** confirmed (Y), **7** partial, **6** to be confirmed (and 7 not implemented), out of 32.

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	[ ]	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	[ ]	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	[ ]	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	[ ]	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	[ ]	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Auto-Set Waste Status on Collection

```
// Script ID : DS-01
// Location : Collection Records → On Add → On Success
// Purpose  : Mark the linked waste record as Collected once a
collection record is saved

// Fetch the waste record linked to the collection record wasteRecord =
agricultural_waste_records[ID == input.waste_record];
// Check if the waste record exists and update its status to Collected
if(!wasteRecord.isNull())
{
    wasteRecord.status="Collected";
}
```

C.2 Validate Collection Request Quantity

```
// Script ID : DS-02
// Location : Waste Collection Requests → On Add/Edit → On Validate
// Purpose  : Block a request whose quantity exceeds the available
waste quantity

// Fetch the waste record referenced in the request using the lookup
field
wasteRecord = agricultural_waste_records[ID == input.waste_record];
// Check if waste record exists
if(wasteRecord.isNull())
{
    alert "Error: Waste record not found for the selected waste.";
    cancel submit;
}
// Get the requested quantity from input
requestedQuantity = input.requested_quantity;
// Get the available quantity from the waste record
availableQuantity = wasteRecord.available_quantity;
// Validate if requested quantity exceeds available quantity
if(requestedQuantity > availableQuantity)
{
    alert "Error: Requested quantity (" +
requestedQuantity.toString() + ") exceeds available waste quantity
(" + availableQuantity.toString() + "). Please adjust your
request.";
    cancel submit;
}
```

C.3 Exclude Product on Rejection

```
// Script ID : DS-03
// Location : Product Quality Tests → On Edit → On Update of Approval
Status
// Purpose  : Zero the saleable stock of a rejected product,
append a rejection note and fail a Started batch
```

```
// Check if the approval status is set to Rejected
if(input.approval_status == "Rejected")
{
    // Fetch the related bio product record using the lookup field
    productRecord = bio_products[ID == input.bio_product.ID];
    // Initialize a list to hold inventory records to update
    inventoryRecords = product_inventory[bio_product.ID ==
input.bio_product.ID];
    if(!inventoryRecords.isEmpty())
    {
        for each invRecord in inventoryRecords
        {
            // Ensure the product is excluded from saleable stock
            // Set available stock to 0 or reduce by the tested
quantity if tracked
            currentAvailable = ifNull(invRecord.available_stock,0);
            invRecord.available_stock=0;
            // Optionally, you could move the quantity to a
'reserved_stock' or 'rejected_stock' field if such fields existed
            // For now, we simply zero out the available stock to
exclude from sale
        }
    }
    // Record the rejection remarks in the product_quality_tests record
    // Using test_parameters to store rejection details as no dedicated
remarks field is specified
    rejectionMessage = "Product Rejected on " + zoho.currentdate;
    if(!input.test_parameters.isNull())
    {
        input.test_parameters = input.test_parameters + "\n" +
rejectionMessage;
    }
    else
    {
        input.test_parameters = rejectionMessage;
    }
    // Update the related processing batch status to 'Failed' if it was
'Started'
    if(!input.processing_batch.isNull())
    {
        batchRecord = processing_batches[ID ==
input.processing_batch.ID];
        if(batchRecord.batch_status == "Started")
        {
            batchRecord.batch_status="Failed";
        }
    }
}
```

C.4 Set Pending Status on New Quality Test

```
// Script ID : DS-04
// Location : Product Quality Tests → On Add → On Load
// Purpose  : Default the test result to Pending for a new test
record
```

```
// Set the test_result field to 'Pending' when the form loads for a new
record
if(input.ID == 0)
{
    input.test_result = "Pending";
}
```

C.5 Make Product Available on Approval

```
// Script ID : DS-05
// Location : Product Quality Tests → On Edit → On Update of Approval
Status
// Purpose    : On approval, confirm an inventory row exists for the
product and centre, creating one if missing

// Check if the approval status is set to 'Approved'
if(input.approval_status == "Approved")
{
    // Fetch the associated bio product record using the lookup field
    productRecord = bio_products[ID == input.bio_product.ID];
    // Fetch the corresponding inventory record for this product
    // Note: processing_center is fetched from the processing_batch
lookup
    processingCenterID = input.processing_batch.processing_center.ID;
    inventoryRecord = product_inventory[bio_product.ID ==
input.bio_product.ID && processing_center.ID == processingCenterID];
    if(inventoryRecord.count() > 0)
    {
        // Update the available stock to make it available for sale
        // Assuming we add the tested quantity or simply ensure
availability
        // Since specific quantity logic wasn't provided, we ensure
the record is active
        info "Product " + productRecord.product_name + " is now available
for sale after quality approval.";
    }
    else
    {
        // If no inventory record exists, create one to make the product
available
        insert into product_inventory
        [
            bio_product=input.bio_product
            processing_center=input.processing_batch.processing_center
            available_stock=0
            last_updated_date=zoho.currentdate
            Added_User=zoho.loginuser
        ]
        info "New inventory record created for product " +
input.bio_product.product_name + " to enable sales.";
    }
}
```

C.6 Update Status on Approval

```
// Script ID : DS-06
```

```
// Location : Waste Collection Requests → On Edit → On Update of
Request Status
// Purpose  : Log that an approved request is ready for collection

// Check if the request status has changed to "Approved"
if(input.request_status == "Approved")
{
    // Update the status to "Ready for Collection" (assuming this is
a valid option or a new field)
    // Note: If "Ready for Collection" is not an existing option in
request_status,
    // you may need to add it to the single_select options or use a
different field.
    // For now, we assume there's a field or we update a related status
indicator.
    // If there is a separate field for collection readiness, update
it:
    // collection_ready = "Yes";
    // If the request_status field itself needs to be updated to a new
value like "Ready for Collection",
    // ensure "Ready for Collection" is added to the single_select
options in the form definition.
    // Otherwise, this code will update the status to "Approved" which
is already the current value.
    // Example: If you want to log or trigger an action when status
becomes Approved
    info "Waste collection request with ID " + input.ID + " has been
approved and is ready for collection.";
    // Optionally, you can create a collection schedule or notify
the collection team here
    // by fetching relevant data and creating records in
collection_schedules form.
}
```

C.7 Set Pending Status on New Request

```
// Script ID : DS-07
// Location : Waste Collection Requests → On Add → On Load
// Purpose  : Default the request status to Pending for a new
request

// Set the request status to Pending when the form loads for a new
record
if(input.ID == 0 || input.ID == null)
{
    input.request_status = "Pending";
}
```

C.8 Validate Product and Calculate Sale Amount

```
// Script ID : DS-08
// Location : Product Orders → On Add/Edit → On Validate
// Purpose  : Check product approval and stock, then compute the
order total (quantity × selling price)

// Validate that the selected bio product exists and is approved
productRecord = bio_products[ID == input.bio_product];
```

```
if(productRecord.isNull())
{
    alert "Selected product is not found."; return;
}
else
{
    // Check if the product has passed quality tests qualityCheck
    = product_quality_tests[bio_product ==
productRecord.ID && approval_status == "Approved"];
    if(qualityCheck.isNull())
    {
        alert "The selected product has not been approved yet.";
        return;
    }
    else
    {
        // Check inventory stock availability inventoryRecord
        = product_inventory[bio_product ==
productRecord.ID];
        if(inventoryRecord.isNull())
        {
            alert "No inventory found for the selected product.";
            return;
        }
        else
        {
            availableStock = inventoryRecord.available_stock;
            requestedQuantity = input.order_quantity;
            if(requestedQuantity > availableStock)
            {
                alert "Insufficient stock. Available: " +
availableStock + ", Requested: " + requestedQuantity;
                return;
            }
            else
            {
                // Calculate total sale amount
                unitPrice = productRecord.selling_price; totalAmount
                = requestedQuantity * unitPrice;
                // Update the order record with calculated total
                input.total_amount = totalAmount;
                // Validation passed, no alert needed
            }
        }
    }
}
```

C.9 Reduce Stock on Sale Confirmation

```
// Script ID : DS-09
// Location : Product Orders → On Add → On Success
// Purpose  : Reduce available stock by the ordered quantity after
the order is saved

// Fetch the product order record that was just submitted
orderRecord = product_orders[ID == input.ID];
// Get the bio product and quantity from the order
selectedProduct = orderRecord.bio_product;
```

```
orderQuantity = orderRecord.order_quantity;
// Check if product and quantity are valid if(!selectedProduct.isNull()
&& !orderQuantity.isNull() && orderQuantity > 0)
{
    // Fetch the current inventory record for this product
    // Note: Adjust criteria if multiple inventory records exist per
product
    inventoryRecord = product_inventory[bio_product ==
selectedProduct];
    // If inventory record exists, reduce the available stock
    if(!inventoryRecord.isNull())
    {
        currentStock = inventoryRecord.available_stock; if(currentStock
        >= orderQuantity)
        {
            newStock = currentStock - orderQuantity;
            inventoryRecord.available_stock=newStock;
            info "Stock updated successfully. New available stock: "
+ newStock;
        }
        else
        {
            info "Error: Insufficient stock. Available: " +
currentStock + ", Required: " + orderQuantity;
        }
    }
    else
    {
        // If no inventory record exists, log error
        info "Warning: No inventory record found for product ID: "
+ selectedProduct;
    }
}
else
{
    info "Error: Invalid product or quantity in order.";
}
```

C.10 Validate Storage Capacity on Create

```
// Script ID : DS-10
// Location : Agricultural Waste Records → On Add/Edit → On Validate
// Purpose  : Intended to check processing-centre capacity before
accepting a waste record

// Fetch the related processing center from the lookup field
'processing_center' in the 'processing_batches' form
// Note: 'agricultural_waste_records' does not have a direct
'processing_center' field.
// Assuming the waste record is linked to a 'processing_batches' record
via 'waste_id' or similar,
// or the user intended to check capacity against a specific processing
center selected elsewhere.
// Since 'processing_center' is not in 'agricultural_waste_records',
we must check the context.
// If this is a validation on 'agricultural_waste_records', and no
processing center is selected yet,
```

```
// we cannot check capacity.
// However, if the user meant to check against a 'processing_center'
that will be linked later,
// or if there is a misunderstanding in the form structure.
// Let's assume the user wants to check capacity for a
'processing_center' that is linked via a 'processing_batches' record
// which might be created later or is linked via a different mechanism.
// But based on the error, 'processing_center' is not in
'agricultural_waste_records'.
// Let's re-examine the form meta:
// agricultural_waste_records has: farmer, farm, waste_category, status,
etc.
// It does NOT have 'processing_center'.
// 'processing_batches' has 'processing_center'.
// If the intention is to check capacity when a 'processing_batch' is
created or linked,
// the script should be on 'processing_batches' form, not
'agricultural_waste_records'.
// However, if the user insists on this script on
'agricultural_waste_records',
// and there is no 'processing_center' field, we must alert the user
or skip.
// Let's assume the user made a mistake and meant to check capacity for
a 'processing_center'
// that is linked via a 'processing_batches' record. But since we are
on 'agricultural_waste_records',
// we cannot access 'processing_center' directly.
// Alternative: If the user meant to check capacity for a
'processing_center' that is selected
// in a different form or context, we need to know how to access it.
// Since the error is clear: 'processing_center' does not exist in
'agricultural_waste_records',
// we will adjust the script to check if the user intended to check
capacity for a 'processing_center'
// that is linked via a 'processing_batches' record. But without a
direct link, we cannot proceed.
// Let's assume the user meant to check capacity for a
'processing_center' that is linked via a 'processing_batches' record
// and that 'processing_batches' is linked to
'agricultural_waste_records' via 'waste_id' or similar.
// But 'processing_batches' has 'collection_record' which links to
'collection_records', which links to 'agricultural_waste_records'.
// Let's try to find the 'processing_batches' record that is linked to
the current 'agricultural_waste_records' record.
// But 'processing_batches' has 'collection_record' which links to
'collection_records', which has 'waste_record' linking to
'agricultural_waste_records'.
// So, we can find the 'processing_batches' record by:
// 1. Finding the 'collection_records' record that has 'waste_record'
equal to the current 'agricultural_waste_records' ID.
// 2. Then finding the 'processing_batches' record that has
'collection_record' equal to the 'collection_records' ID.
// But this is complex and might not be what the user intended.
// Let's assume the user made a mistake and meant to check capacity for
a 'processing_center' that is selected
```

```
// in the 'processing_batches' form. In that case, the script should
be on 'processing_batches' form.
// Since we are on 'agricultural_waste_records' and there is no
'processing_center' field,
// we will alert the user that the field does not exist and skip the
check.
alert "The field 'processing_center' does not exist in the
'agricultural_waste_records' form. Please check the form
structure.";
cancel submit;
```

C.11 Show Low Stock Alert

```
// Script ID : DS-11
// Location : Product Inventory → On Add/Edit → On Validate
// Purpose  : Warn the user when available stock is below the
reorder level

if(input.available_stock < input.reorder_level)
{
    alert "Low Stock Alert: The available stock for this product is
below the reorder level. Please place a new order.";
}
```

C.12 Monthly Inventory Reorder Alert

```
// Script ID : DS-12
// Location : Schedule on Product Inventory (10:00 IST,
Asia/Kolkata)
// Purpose  : E-mail the administrator an HTML low-stock notice
for each product below reorder level

// Check if available stock is below reorder level
if(input.available_stock < input.reorder_level)
{
    // Prepare email subject
    subject = "Low Stock Alert: " + input.bio_product.product_name;
    // Prepare email message
    message = "<p>Dear Team,</p>" + "<p>The following product stock has
fallen below the reorder level:</p>" + "<ul>" +
"<li><strong>Product:</strong> " + input.bio_product.product_name +
"</li>" + "<li><strong>Available Stock:</strong> " +
input.available_stock.toString() + "</li>" + "<li><strong>Reorder
Level:</strong> " + input.reorder_level.toString() + "</li>" +
"<li><strong>Location Code:</strong> " + input.location_code + "</li>"
+ "<li><strong>Last Updated:</strong> " +
input.last_updated_date.toString() + "</li>" + "</ul>" + "<p>Please
take necessary action to replenish the stock.</p>";
    // Send email alert
    sendmail
    [
        from :zoho.loginuserid
        to :zoho.adminuserid
        subject :subject
        message :message
        content type :HTML
    ]
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Open the published application link [https://creatorapp.zoho.in/...] in a browser and sign in with the Zoho account created for your profile; portal users (farmers and buyers) use the portal sign-in link sent to them by e-mail. On a phone, install the Zoho Creator app, sign in with the same account and open GreenAgro BioLoop. The left menu lists the forms and reports you are allowed to use; anything not listed is outside your profile.

D.2 Instructions for Administrator

- Open Dashboard to view total order amount, average efficiency, waste received and reused, and waste-status counts.
- Add or edit users and assign a permission profile in the application's Users settings.
- Open any list or kanban report; click a record to view its quick view, or Edit to change it.
- Activate the Monthly Inventory Reorder Alert schedule so that low-stock e-mails are received.

D.3 Instructions for Waste Collection Coordinator

- Open Waste Collection Requests and review requests by priority; move each through the Blueprint (Under Review → Approved → Scheduling).
- Open Collection Schedules and add a schedule: choose the request, team, date, time and route.
- After the pickup, add a Collection Record with collected quantity, actual weight and quality check; the waste record turns Collected automatically.

D.4 Instructions for Processing Center Manager

- Open Processing Batches and add a batch: select the collection record, centre and method, and enter the input quantity and dates.
- Move the batch through its Blueprint stages from Batch Initiated to Batch Closed.
- Add a Processing Record with output, residue, efficiency, grade and energy; log the Product Quality Test for the batch.

D.5 Instructions for Environmental Compliance Officer

- Open Environmental Impact Records and add a record per batch: waste received, waste reused, CO_2, water and land saved.
- Set Verification Status to Verified after checking the figures.
- Review Product Quality Tests and their kanban by test result.

D.6 Instructions for Inventory and Distribution Specialist

- Open Product Inventory and update available stock, reserved stock and reorder level; a warning appears if stock is below the reorder level.
- Open Product Orders to review orders; open Product Distribution and record dispatch and delivery details.
- Maintain the Bio Products catalogue (price, unit, shelf life, certification).

D.7 Instructions for Farmer and Bio-Product Buyer (Portal)

- Sign in to the portal; farmers can update farm details and raise a Waste Collection Request.
- Open Product Orders and add an order: choose the product and quantity; the total is calculated for you.
- Use the reports to follow the status of your requests and orders.

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	[Student Name]	**Register No.**	[Register Number]
Project Title	GreenAgro BioLoop	**Date**	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner Name:	Signature of External Examiner Name:

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	[Student Name]	**Register No.**	[Register Number]
Department / Section	[ ]	**Date of Viva**	
Application URL Verified	Yes / No	**Demo Duration**	*m*

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

INTERNAL EXAMINER **Name:**	**EXTERNAL EXAMINER** **Name:**